Deriving clause incorporation in Inuktitut*

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1 Introduction

Although most prior work on the morphosyntax of incorporation has focused on the incorporation of (typically bare) nouns into verb complexes, it is also known that some languages also permit the incorporation of complex verbal or even clausal constituents (Baker, 1988; Pittman, 2006, 2009; Wojdak, 2008; Compton and Pittman, 2010; Barrie and Mathieu, 2016; McKenzie, 2019, 2022; Olthof, 2020a,b; Yuan, 2025, a.o.). The incorporation of the latter raises questions for the syntax-morphology interface that remain largely unexplored. To what extent do incorporated clauses differ syntactically from their standalone counterparts? How might existing analyses of noun incorporation be extended to the incorporation of complex constituents that contain their own nominal arguments? And what does the possibility of clause incorporation reveal about the morphosyntactic underpinnings of complex word-formation more broadly?

As a window into these questions, this paper investigates the morphosyntax of a type of embedded incorporated clause in (Eastern Canadian) Inuktitut and related languages, found in a construction I refer to as the *clause-incorporating existential*. As shown in (1a), a verb bearing so-called 'participial' clause type morphology is incorporated into another verb *-qaq* 'have', which functions here as a verb of existence (incorporating verbs are underlined throughout the paper for clarity). The existential pivot, interpreted as the subject of the embedded verb, is stranded and marked with so-called 'modalis' (MOD) case. The incorporated clause displays both similarities and differences with the standard declarative construction, (1b), which bears surface-similar clause type morphology but contains an ABS subject and lacks the indefinite reading found in (1a).

(1) **Incorporated vs. standalone "participial" clauses in Inuktitut**

a. anguti-mik tikit-tu-qaq-tuq
man-MOD arrive-PART-have-PART.3SG.S
'A man arrived.'

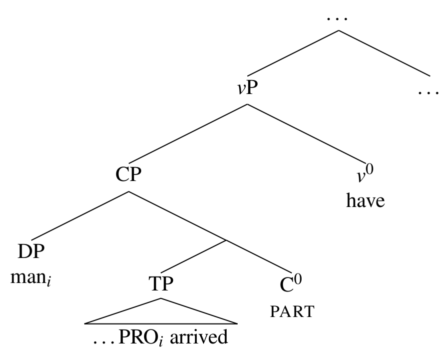
b. anguti tikit-tuq
man.ABS arrive-PART.3SG.S
'The man arrived.'

Although the clause-incorporating existential appears frequently in documentary sources and corpora as a strategy to express indefinite subjects, it has not been systematically investigated, as previous work on incorporation in the Inuit-Yupik languages has mainly centered around that of nominal direct objects (Sadock 1980, 1986, 1991; Woodbury and Sadock 1986; van Geenhoven 1998, 2002; Johns 1999, 2007b, 2009; Compton and Pittman 2010; Woodbury 2017; Yuan 2025, a.o., though see Pittman 2006, 2009 on incorporated verbal constituents). The primary goals of this paper are therefore to uncover the morphosyntactic properties of this construction, and to connect these properties to the underpinnings of incorporation and complex (polysynthetic) word-formation in Inuktitut more generally. Additionally, we will see that the clause-incorporating existential offers a new empirical window into the case and agreement system of the language, as the nominal arguments introduced within incorporated clauses do not display the same morphosyntactic patternings as their counterparts in non-incorporated clauses.

*Acknowledgments will be included in a later version of this paper.

I propose that the incorporated clause in (1a) is a CP in category, but it lacks higher the CP-level ϕ-probes present in root declarative clauses. This bleeds ERG and ABS case assignment to DPs in that clause, providing novel evidence that case assignment is specifically keyed to the extended CP domain (e.g. Yuan, 2018, 2022; Carrier and Compton, to appear), rather than lower projections such as TP and *v*P. Moreover, I argue that the clause is a *pseudorelative*, that is, a clause-sized predicate of a small clause, related to its subject by control (e.g. Graffi, 1980, 2017; Cinque, 1995; Angelopoulos, 2015; Moulton and Grillo, 2015). Although previous work on pseudorelatives (mainly on European languages) has generally focused on their usages in perception verb constructions rather than existentials, my proposal is motivated by strong empirical parallels between these construction types both in Inuktitut and cross-linguistically. Just like pseudorelatives in other languages, the embedded clause in the Inuktitut clause-incorporating existential is tense-deficient, resembles (yet is syntactically distinguishable from) relative clauses, and displays a clause-internal subject gap restriction suggestive of control. Put together, (2) provides a simplified schematization of the analysis I develop. Note that (2) assumes Matushansky's (2019) treatment of small clauses, in that the subject is represented as the specifier of the clausal predicate.

(2) **Pseudorelative structure in Inuktitut (preliminary)**



In (2), the subject and the embedded clause are both within the scope of the incorporating verb 'have' (*-qaq*), yet it is the clause that gets incorporated in this construction. I argue that this follows straightforwardly from a postsyntactic account of incorporation that is nonetheless keyed to syntactic structure, as recently developed by Yuan 2025 (who builds on Sadock 1985, 1991 and Bok-Bennema and Groos 1988, in turn). Yuan's analysis invokes the operation Merger, which creates complex words out of structurally adjacent heads (e.g. Marantz, 1988; Embick and Noyer, 2001). Importantly, the particular subtype of Merger involved permits an element to undergo Merger only with the head of its *complement*; heads of specifier and adjunct XPs within that element's c-command domain are not possible targets. Applying this to (2), v^0 may not undergo Merger into the nominal subject, but it may do so iteratively down the clausal spine.

Finally, this analysis has consequences for the nature of complex word-formation in Inuktitut. Compton and Pittman 2010 offer an influential theory of polysynthetic wordhood in the Inuit languages, arguing that clausal and nominal syntactic phases (i.e., CPs and DPs) map to individual phonological words, while smaller non-phasal constituents only ever correspond to subword material. Given this strict syntax-phonology mapping, Compton and Pittman's theory dispenses with the idea that individual syntactic elements are lexically specified as morphologically free vs. affixal. However, the clause-incorporating existential provides an empirical challenge to this theory. Although the embedded clause is reduced, lacking its highest syntactic layers, I argue that that is *not* what triggers its incorporated status—pseudorelative clauses may actually surface both as incorporated and as independent words in different syntactic contexts. Moreover, we will see that non-reduced embedded clauses may likewise be standalone or incorporated, determined solely by the morphological properties of the embedding verb. I conclude that complex word-formation in Inuktitut is determined by the morphological subcategorization properties of individual heads (i.e., affixal or not), as also argued by Sadock 1985, 1991, and that word formation via head-head adjacency may in fact be sufficient to capture even polysynthetic languages such as Inuktitut.

The remainder of this paper is organized as follows. In §2, I overview key properties of Inuktitut morphosyntax relevant to the paper. §§3-4 then provide a detailed investigation of the clause-incorporating existential: after detailing the grammatical properties of the embedded clause and its nominal arguments (§3), I argue that the incorporated clause is a pseudorelative (small clause) predicate embedded under an affixal verb of existence (§4). §5 then provides an account of case assignment in the clause-incorporating existential, building on the findings of §§3-4. Finally, §6 extends Yuan's (2025) postsyntactic analysis of noun incorporation to the clause-incorporating existential and explores implications of this system for treatments of polysynthetic word-formation.

2 Overview of Inuktitut morphosyntax

I start by presenting key properties of Inuktitut and Inuit, focusing on its case and agreement system, the functions of participial clause type morphology, and the incorporation of nominal and verbal constituents.

2.1 Language background and key morphosyntactic properties

The Inuit languages (of the Inuit-Yupik-Unangan language family) are a continuum of dialects spanning the North American Arctic and Greenland. This paper focuses primarily on the varieties of the Inuktitut language spoken in the Qikiqtaaluk (Baffin Island) region of Nunavut,[1] although data from other varieties are provided wherever relevant. Unless explicitly indicated, uncited examples were elicited by the author between 2017-2019, in Iqaluit, Nunavut and remotely (online).[2]

Like other languages of the family, Inuktitut is considered polysynthetic, with a large number of derivational suffixes that encode verbs, adjectives, and adverbs, (3) (e.g. Fortescue, 2002, 2017; Cook and Johns, 2009; Compton, 2012; Compton and Bliss, 2023).

(3) **Complex 'polysynthetic' word in Inuktitut**
annulaksi-kkanni-nginna-jualu-gasu-lauqsima-guma-nngit-tsiaq-galuaq-tunga
imprison-again-really-a.lot-try-ever-want-NEG-EMPH-EMPH-PART.1SG.S
'I would never ever even want to try to end up in jail ever again even for a bit.' (NB; Johns 2007b, p. 564)

The language has been described as having base SOV word order, although SVO is also frequently attested in neutral contexts, possibly due to language contact with English (Fortescue, 1993). Other word orders are also permitted, which I take to arise from scrambling. I assume that the language is head-final and adheres to the Mirror Principle, so word-internal morpheme order corresponds to syntactic height (Baker, 1985, et seq.). A complex verb begins with a root, which is (optionally) followed by various VP- and TP-level suffixes, and ends with clause type morphology, often also referred to as *mood* or *modality* in the Inuit literature (e.g. Fortescue, 1983; Dorais, 1988).[3] The examples in (4) also show that verbal agreement is morphologically sensitive to clause type. Thus, both subject and object agreement is located in the CP-domain, a point that will

[1] A broader term, Inuktut, is used by the Government of Nunavut to encompass Inuktitut and Inuinnaqtun.

[2] Following the conventions of Yuan 2025, this paper indicates for each linguistic example the community of the Inuktitut speaker that produced it. The following abbreviations will be used for examples elicited by the author: AB = Arctic Bay (Ikpiarjuk), IG = Iglulik, PG = Pangnirtung (Pangniqtuuq), PI = Pond Inlet (Mittimatalik), IQ = Iqaluit, SN = Sanirajak. For examples cited from published sources on Inuktitut and other languages, the following additional abbreviations, for both individual communities and broader regions are used, if the information is available: CH = Coral Harbour (Salliq), IT = Itivimiut, KL = Kalaallisut, L = Labrador, NB = North Baffin, SB = South Baffin, TR = Tarramiutut.

[3] However, this usage of the terms 'mood' and 'modality' differs from how it is typically used outside of Inuit linguistics. Thus, I continue to refer to the relevant distinctions in terms of clause type, following Compton and Yuan to appear.

be reinforced below as well as throughout the paper (e.g. Johns, 2007b; Compton, 2016, 2018b; Yuan, 2021, 2022; Carrier and Compton, to appear; Compton and Yuan, to appear).

(4) **Clause type and agreement morphology**

a. nirua-runna-**tait**
choose-can-PART.2SG.S/3SG.O
'You can pick it.' (AB)

b. ani-qu-api-**guk**
leave-tell-small/dear-IMP.2SG.S/3SG.O
'Tell the dear one to leave.' (TR; Beach 2011, pp. 167)

Although these agreement forms are often realized as morphologically unsegmentable portmanteaux, they are comprised of distinct syntactic heads, which I take to be subprojections of an articulated CP domain (e.g. Rizzi, 1997). Certain suffixes may intervene between clause type and agreement morphology, (5a);[4] this additionally reveals that ϕ-agreement is located syntactically above clause type morphology. Moreover, certain combinations transparently show object ϕ-morphology realized outside of the portmanteau forms (now comprising just clause type and subject agreement), as exemplified by (5b), allowing us to conclude that ϕ-morphology indexing objects is higher than that indexing subjects.

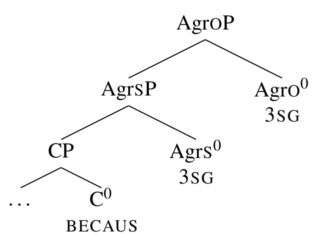
(5) **Clause type < subject ϕ-morphology < object ϕ-morphology**

a. numaa-suk-**tu**-inna-**it**
sad-feel-PART-every-3PL.S
'All of them are sad.' (AB)

b. taku-mmat vs. taku-**mma**-**uk**
see-BECAUS.3SG.S see-BECAUS.3SG.S-3SG.O
'because (s)he/it sees' vs. 'because (s)he/it sees it' (Yuan, 2022, p. 514)

Loosely adapting the conclusions of Yuan (2022), this suggests a representation as in (6), which illustrates (5b). The clause type morphology expones the head I label 'C' for convenience; the labels 'AgrS' and 'AgrO' are used here purely to indicate which heads Agree with which nominal arguments.

(6) **Extended CP-domain in Inuktitut based on (5b)**



As seen in (1), the embedded clause within the clause-incorporating existential bears so-called 'participial' morphology. Root declarative clauses in Inuktitut also often surface with this morphology, rather than the 'indicative' clause type morphology standardly used to mark declaratives in other varieties such as Kalaallisut (in Inuktitut, the indicative is somewhat pragmatically marked).[5] Participial morphology is *t*- or *j*-initial, depending on whether the stem to which it attaches ends in a consonant or vowel, (7). Throughout this paper, intransitive and transitive verbs bearing participial clause type morphology will be referred to as 'participial clauses' as shorthand.

[4] On the morphological integration of noun-modifying suffixes into the verb complex, see Snigaroff 2024 on distantly related Aleut.
[5] And in other varieties (e.g., Kangiryuarmiut, a Inuinnaqtun variety), the indicative is absent altogether (Lowe, 1985).

(7) **Phonologically-conditioned allomorphy in participial morphology**

a. malik-**t**unga vs. taku-**j**unga
follow-PART.1SG.S see-PART.1SG.S
'I follow' vs. 'I see'

b. malik-**t**agit vs. taku-**j**agit
follow-PART.1SG.S/2SG.O see-PART.1SG.S/2SG.O
'I follow you' vs. 'I see you'

Verbs bearing intransitive and transitive participial morphology are generally considered to have been historically derived from *nominal participles* (e.g. Fortescue et al., 2011), hence the terminology. However, it is debatable to what extent this morphology is *synchronically* nominal. The position I take in this paper is that participial morphology is now uniformly part of the verbal extended sequence, having been fully integrated into the complementizer system of the language (see also Johns 2000; Mahieu 2009; Compton 2018a, 2019).[6] I moreover assume that this holds for both root declarative clauses and embedded participial clauses: in all contexts, participial morphology encodes C^0.

Inuktitut transitive sentences display an ergative (ERG-ABS) case alignment, though the ergative construction seen in (8a) alternates with an antipassive (ABS-MOD) construction, (8b). Notably, antipassive objects in Inuktitut and other Eastern Canadian varieties are *not* necessarily interpreted as non-specific or indefinite (contrary to how they are typically described cross-linguistically), and do not reflect an argument structural change such as detransitivization (e.g. Johns 1999, 2001, 2006, 2017; Beach 2011; Carrier 2017, 2020; Yuan 2018, 2021, 2022, cf. Baker 1988; Wharram 2003).

(8) **Ergative and antipassive case and agreement patternings**

a. Taiviti-**up** **igalaaq** sura-qqau-janga
David-ERG window.ABS break-REC.PST-PART.3SG.S/3SG.O
'David broke the window (earlier today).'

b. **Taiviti** igalaar-**mik** surak-si-qqau-juq
David.ABS window-MOD break-AP-REC.PST-PART.3SG.S
'David broke a/the window (earlier today).' (AB)

[6] In contrast, Johns 1992 proposes that participial intransitive and transitive declaratives are synchronically null copular constructions whose predicates are nominalized participial verbs. See also Yuan 2013 and Carrier and Compton to appear, building on Johns, as well as the summary in Compton 2017c. Under this analysis, sentences like (ia-b) would have the alternative parses given below, where the ABS argument is invariably the subject. Note that the transitive subject in (ib) would be reinterpreted as a possessor of the nominal passive verb. The latter is supported by the homophony between ERG and GEN case (*-up*) and between subject/object participial agreement and possessor/possessee agreement (cf., *qimmi-nga* 'his/her dog').

(i) **Alternative parses for intransitive and transitive participial constructions**

a. (*pro*) mumiq-tuq
(3SG.PRON.ABS) dance-PART
'S/he dances.' (Or: 'S/he is a dancing one.')

b. Taiviti-up igalaaq sura-qqau-ja-nga
David-GEN window.ABS break-REC.PST-PASS.PART-3SG/3SG
'David broke the window (earlier today).' (Or: 'The window is David's earlier today broken one.')

However, a major challenge for this proposal as a synchronic state of affairs is that it cannot extend to participial clauses containing 1st/2nd person ABS arguments (e.g., *taku-ju-nga* 'I see' or *taku-ja-git* 'I see you' in (7)). This is because the agreement forms cross-referencing such arguments do not have nominal analogues (e.g., possessees are inherently 3rd person); see also Carrier and Compton to appear for additional morphosyntactic differences (in response to this, Carrier and Compton to appear propose a syntactic split, whereby only the participial clauses containing 1st/2nd person ABS arguments are verbal). Therefore, I adopt a simpler picture, whereby participial clauses not syntactically different from any other clause type. Following Mahieu 2009, I assume that the analysis above represents just the *diachronic* pathway of participial clauses but not their synchronic status, the originally nominal morphology having been reanalyzed as complementizers.

Up to two nominals may be indexed by ϕ-morphology, and these nominals are necessarily ERG and ABS, respectively; as seen in constructions with three (or more) non-oblique arguments, (9), other nominals in the clause are not cross-referenced by ϕ-agreement and are either MOD or oblique.[7]

(9) **Double and triple object constructions in Inuktitut**

a. Miali-**up** **Jaani** tuni-qqau-**janga** ajjinnguar-$\boxed{\text{mik}}$
Miali-ERG Jaani.ABS give-REC.PST-PART.3SG.S/3SG.O picture-MOD
'Miali gave Jaani a picture.' (IG)

b. **(*pro*)** **Jaani** qiturnga-$\boxed{\text{nik}}$ tuni-si-qu-**jara** mamaq-tu-$\boxed{\text{nik}}$
(1SG.ERG) Jaani.ABS offspring-PL.MOD give-AP-want-PART.1SG.S/3SG.O tasty-PART-PL.MOD
'I want Jaani to give the children candy (lit. tasty things).' (IG)

Given these facts, this paper will defend the generalizations about Inuktitut case and agreement summarized in (10). The morphosyntactic properties observed in the clause incorporation existential will provide novel evidence for these points.

(10) **Case and agreement in Inuktitut are tied to the extended CP domain**

a. ERG and ABS cases (indirectly) reflect successful Agree with CP-level ϕ-probes, while MOD case reflects no such ϕ-Agree relations

b. No other functional heads (e.g., located in the TP- and *v*P-domains) are implicated in case assignment and ϕ-agreement

These generalizations diverge from most other accounts of ergative case and agreement systems in the literature, which make use of lower heads. For instance, ERG case is often argued to be assigned by v^0 (Woolford, 2006; Aldridge, 2008; Legate, 2008; Coon et al., 2021; Tollan and Massam, 2022; Clem and Deal, to appear, a.m.o.), by T^0 (Bobaljik, 1993; Rezac et al., 2014), or through a combination of both heads (Deal, 2010; Clem, 2019; Ganenkov, 2022); this literature also often takes ABS to be assigned by v^0 or T^0. Likewise, it is canonical for ϕ-agreement with subjects and objects to be analyzed as located in T^0 and/or v^0.

However, we will see that the distributions of ERG and ABS in Inuktitut are not determined by these heads, but instead require the CP-level heads shown in (6). In §5, I develop an account in which ERG and ABS case assignment takes place within the syntactic domain associated with AgrOP, the highest projection of the clause, while MOD case assignment is assigned within *v*P. Leaving the details of such an analysis until then, for now it suffices to remember that the distributions of ERG, ABS, and MOD are tied to syntactic height.

Finally, although only CP-level heads are implicated in agreement and case, I still assume that T^0 is still syntactically active in that it has an EPP requirement, satisfied by either moving a nominal to Spec-TP or Merging an expletive there. This is relevant to the analysis of the clause-incorporating existential construction, as I show in §§3-5. The idea that ABS and MOD case assignment is tied to nominal height is also relevant to embedded subjects. Although subjects usually surface *v*P-externally in Spec-TP, in the clause incorporation construction they appear within *v*P, under the verb *-qaq* 'have'. As already shown in (1) above, such subjects are MOD, not ABS, an initial piece of support for the generalizations in (10).

2.2 Incorporation and arguments of incorporated elements

In general, noun incorporation in the world's languages tends to be optional for any given transitive verb, its occurrence often correlating with a non-referential interpretation of the nominal object. In contrast, as shown in (11), noun incorporation in Inuktitut (as well as in the Inuit-Yupik languages more broadly) is obligatory with a small class of verbs, and impossible with all other verbs. Moreover, as discussed in Yuan 2018, 2025, Inuktitut

[7] Though see Johns 2018, which documents the ability for MOD objects to be indexed by object agreement for some Labrador Inuttut speakers.

incorporated objects do not obligatorily display the aforementioned interpretive restrictions.[8]

(11) **Noun incorporation is obligatory with affixal verbs**

a. **pitsi**-tu-vunga
dried.fish-consume-IND.1SG.S
'I'm eating dried fish.' (cf. *pitsik* 'dried fish')

b. ***pitsi-mik** tu-vunga
dried.fish-MOD consume-IND.1SG.S
Intended: 'I'm eating dried fish.' (L; Johns 2007b, p. 541)

That incorporation has taken place in the examples throughout this paper is signaled by regular morphophonological changes that affect the final segment of the stem (e.g. Dorais, 1985). Additionally, incorporated elements generally do not bear case or ϕ-features. As pointed out by both Sadock (1985, 1991) and Yuan (2025), however, this does not mean that incorporated elements lack case and number in the syntax; rather, both authors take these features to be syntactically present but simply unrealized due to the morphological subcategorization requirements in incorporation contexts. Indeed, the underlying case and number properties of incorporated nominals may be indicated by case concord with stranded modifiers. In (12), the appearance of the plural MOD form *-nik* on *pingasut* 'three' suggests that the incorporated object is also MOD and plural.[9]

(12) **Case concord on modifiers of incorporated nouns**

ataatsi-**nik** qamuti-qar-poq
one-PL.MOD carriage-have-IND.3SG.S
'He has one carriage.' (KL; Sadock 1980, p. 309)

The verbs that trigger incorporation are often called *affixal verbs* (or *denominal verbs*) in the literature, since they require a morphological host such as the nominal object (Mithun, 1986; Sadock, 1986, 1991; Compton and Pittman, 2010). In general, affixal verbs in the Inuit languages tend to be somewhat semantically bleached compared to their standalone counterparts (Mithun, 1999; Johns, 2007b; Cook and Johns, 2009), though some exceptions have been recently noted by Huijting 2024. In this paper, I follow Johns 2007b, 2009 in differentiating affixal (incorporating) and non-affixal (non-incorporating) verbs as v^0 and V^0, respectively, though this is not of analytical importance.

Incorporated elements in Inuktitut are underlyingly *syntactically phrasal*, even though they are ultimately realized morphologically as complex heads. As shown in (13), they may be internally complex (Compton, 2013). In (13a), the incorporated noun bears several adjectival suffixes. Note also that the affixal verb here is *-qaq* 'have', encoding possession rather than existence and incorporating a nominal object rather than a participial clause. Additionally, the incorporated noun can be shown to be a DP in category; in (13b-c), the objects are interpreted as referential despite being incorporated.

(13) **Incorporation of complex nominals and DPs**

a. **iglu-tsiava-nngua**-qaq-tuq
house-great-pretend-have-PART.3SG.S
'(S)he has a great pretend house.' (SB; Compton 2013, p. 3)

[8] In addition to the Inuit languages, other languages that have been reported to display obligatory noun incorporation with affixal verbs (possibly co-existing with the more 'standard' pattern of optional incorporation) include Halkomelem (Salishan; Gerdts and Hukari 2008), Nuu-chah-nulth and K^{w}ak'^{w}ala (both Wakashan; Waldie 2004; Wojdak 2008; Sardinha 2017), Ojibwe (Algonquian; Mathieu 2013; Barrie and Mathieu 2016), and Chukchi and Koryak (both Chukotko-Kamchatkan; Kurebito 2001, 2017; Kantarovich 2024), though there may be cross-linguistic differences concerning the productivity of the phenomenon.

[9] Sadock (1980) additionally notes that (i) is particularly informative because it shows plural number concord with a (semantically singular) *pluralia tantum* noun.

b. Qallupilluq **Miali**-<u>tu</u>-niaq-pa?
Qallupilluq.ABS Miali-consume-NR.FUT-INT.3SG.S
'Is Qallupilluq going to eat Miali?' (SB; Johns 2009, p. 191)

c. Guuti **uvanga**-<u>liu</u>-lauq-tuq
God.ABS 1SG.PRON-make-PST-PART.3SG.S
'God made me.' (AB)

Another important property of Inuktitut incorporation is that certain affixal verbs embed *complex verbal constituents* rather than nominal ones. Previous authors have analyzed such constructions as involving restructuring or ECM (Pittman, 2006, 2009; Compton and Pittman, 2010; Yuan, 2018). Thus, Inuktitut generally permits the incorporation of clause-like material, in constructions beyond the clause-incorporating existential. In (14), the affixal verb *-niraq* 'say' syntactically embeds a TP clause, as shown by the co-occurrence of matrix and embedded tense morphology in (14a). The example in (14b) additionally shows that a sentential idiom may be embedded under an affixal verb without losing its idiomatic meaning; this reveals that the embedded nominal arguments of such ECM constructions truly originate within the lower clause, under the matrix affixal verb (that is, these arguments are not base-generated in the matrix clause as proleptic arguments).

(14) **Finite TPs embedded under affixal verb *-niraq* 'say'**

a. Saila-up [$_{TP}$ niri-**niar**]-<u>nira</u>-**qqau**-janga [$_{TP}$ Alana
Saila-ERG eat-NR.FUT-say-REC.PST-PART.3SG.S/3SG.O Alana.ABS
tuktu-viniq-mit]
caribou-former-MOD
'Saila said [$_{TP}$ Alana will be eating caribou meat].' (SB; Compton 2017a, p. 299)

b. [$_{TP}$ **tulugait** **qakuq-si-lir**]-<u>niraq</u>-tangit
crow.PL.ABS white-INCP-PROG -say-PART.3SG.S/3PL.O
'She said that [$_{TP}$ Hell is freezing over].'
Lit.: 'She said that [$_{TP}$ the crows are starting to turn white].' (PG)[10]

In (14), only the heads along the embedded functional spine end up morphologically incorporated into the matrix affixal verb; the nominals thematically associated with the embedded verb are not incorporated. The ERG-ABS(-MOD) case arrays seen in (14) are identical to those of transitive and double object constructions, with the ABS arguments (the embedded subjects) indexed by matrix object ϕ-agreement. The morphosyntactic assumptions from §2.1 thus extend to ECM constructions: due to the truncated (non-phasal) nature of the embedded TP, the CP-level ϕ-probe in the matrix clause is able to target the embedded subject in Spec-TP.[11]

Lastly, Inuktitut permits what I call 'recursive' incorporation, arising whenever an affixal verb embeds a constituent containing another affixal verb. Two examples of this are given below. In (15a), verbal constituent consisting of an affixal verb *-liuq* 'make' and its object is nominalized by the suffix *-utaq* 'tool for VERB-ing'. This nominalized constituent then serves as the complement of the affixal verb *-u* 'be'. In (15b), another ECM construction, the embedded TP containing a noun-incorporating affixal verb *-tuq* 'consume' is again

[10] As Yuan 2018 clarifies, the idiomatic expression in the embedded clause in (14b) is a permutation of the actual saying given in (i) (this may be akin to English, 'when pigs fly' vs. 'pigs are flying').

(i) **Original version of idiomatic expression in (14b)**
tulugait qakuq-sip-patta
crow.PL.ABS white-INCP-COND.3P.S
'When Hell freezes over.'
Lit.: 'Only when the crows turn white.' (PG)

[11] So far, this ignores the possibility of a *v*P-level clause-medial phase that the embedded subject must move to in order to be accessible to higher ϕ-probes. This will be revisited in §5.

incorporated into the affixal verb -*niraq* ‘say’.

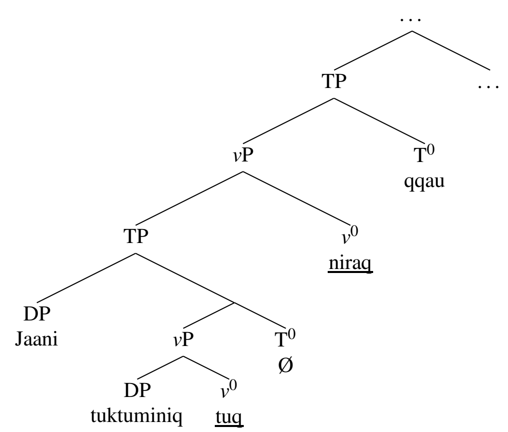
(15) **Recursive incorporation constructions**

a. [$_{nP}$ kiinauja-liur-uta]-u-gunnaq-tut
money-make-tool.for -be-can-PART.3PL.S
'They can make money out of [investments].'
Lit.: '[Investments] can be tools for making money.' (SB; Compton 2012, p. 42)

b. Jamesie-up [$_{TP}$ tuktu-mini-tu]-nira-qqau-janga [$_{TP}$ Jaani]
Jamesie-ERG caribou-former-consume -say-REC.PST-PART.3SG.S/3SG.O Jaani.ABS
'Jamesie said that Jaani ate caribou meat.' (IQ)

To illustrate the syntactic assumptions laid out so far, a simplified and partial structure of (15b) is given in (16) (setting aside issues of SVO word order, as mentioned above). Again, what is crucial here is the idea that affixal verbs (v^0s) may syntactically embed complex constituents—including, potentially, reduced clauses containing multiple nominal arguments.

(16) **Partial structure of incorporation construction**



In sum, this section has discussed the following properties of Inuktitut morphosyntax:

- Participial clause type morphology instantiates C^0 in both root and embedded contexts.
- Case and ϕ-agreement patternings correspond to the presence or absence of Agree dependencies initiated by ϕ-probes in the extended CP-domain, above C^0. ERG and ABS arguments are Agreed with, while MOD arguments are not.
- Incorporation is driven by the affixal nature of certain verbs in the language, in that they require morphological hosts.
- Affixal verbs syntactically embed complex (phrasal) constituents, regardless of whether the individual elements within these constituents are ultimately realized as morphologically free or incorporated.

3 An initial syntax for the clause-incorporating existential

We now turn to the clause-incorporating existential, whose morphosyntactic profile both follows from and further reinforces the properties of Inuktitut detailed above. Although there is no prior literature focusing on this construction, it is frequently attested in documentary sources, corpora, and texts.[12] Therefore, in presenting the

[12] Some examples are provided in (i):

(i) **Examples of clause-incorporating existentials from grammars and texts** first systematic description and analysis of the clause-incorporating existential, this paper provides a blueprint for subsequent work on this and related topics. In this section, I start by introducing the construction and providing a basic syntax of the embedded clause. §4 then details how the embedded clause and the MOD-marked nominal syntactically relate to each other.

3.1 Key morphosyntactic properties

The clause-incorporating existential is not the only way to express existence in Inuktitut, though to my knowledge all other strategies also involve a morpheme that abstractly encodes 'HAVE'. To appreciate the morphosyntax of this particular construction, repeated in (17a), we may contrast it with another existential construction with a more familiar noun incorporation profile, as in (17b). In the latter, the existential affixal verb *-taqaq* (visibly containing *-qaq* 'have') incorporates its nominal complement, and the participial clause associated with the noun is stranded.[13] The two sentences are not translated identically by speakers. Whereas the clause-incorporating existential in (17a) typically gets translated as a simple sentence with an indefinite subject, (17b) is typically translated as a 'there'-existential, with the participial clause serving as a relative clause modifying the indefinite nominal (relative clauses in Inuktitut also bear participial clause type morphology; see §4.2). Put differently, these constructions differ in the locus of the *pivot* of the existential construction, the nominal whose existence is under discussion: in (17a) it is the (standalone) embedded subject and in (17b) it is the incorporated object.

(17) **Existential constructions in Inuktitut**

a. **anguti-mik tikit-tu**-qaq-tuq
man-MOD arrive-PART-have-PART.3SG.S
Speaker's translation: 'A man arrived.'

b. **tikit-tur-mik** **anguti**-taqaq-tuq
arrive-PART-MOD man-exist-PART.3SG.S
Speaker's translation: 'There is a man who arrived.' (AB)

That the subject in (17a) is indeed the existential pivot is demonstrated by its sensitivity to the Definiteness Restriction (Milsark, 1974, et seq.). Proper names are permitted only if interpreted as non-referential, and other definite expressions such as possessive DPs are disallowed entirely, (18a-b). Similarly, when the embedded subject is a null pronoun, it is only understood by speakers as indefinite, (18c).[14] (In possessive constructions,

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a. TV-mi **su-mik** **suqutiginar-tu**-qa-nngil-aq
TV-LOC what-MOD be.interesting-PART-have-NEG-IND.3SG.S
'There is nothing interesting on TV.' (KL; Fortescue 1984, p. 138)

b. nuvuja-lir-tu-alu-u-lir-mat immaqaa pirsi-nia-lir-tuq ullumi **ulluria-mik**
cloud-PROG-PART-big-be-PROG-CAUS.3SG.S perhaps blizzard-NR.FUT-PROG-PART.3SG.S today star-MOD
nuita-ju-qa-nngi-mat
be.visible-PART-have-NEG-CAUS.3SG.S
'It's very cloudy, no stars at all... there may be a snowstorm.' (Patsauq 2021, p. 65; glosses from Mahieu 2021)

[13] Finally, there is an additional existential construction provided in (i) for completeness. Here, the suffix *-lik* is a nominal suffix that roughly has the meaning 'one that has'. I do not discuss this construction further due to paucity of data, but see Yuan 2018, pp. 238-240 on other constructions containing this suffix.

(i) **Existential construction with nominal suffix *-lik***
arnar-nik marruu-nik ani-ju-**lik**
woman-MOD.PL two-MOD.PL leave-PART-have.NMLZ
'Two women left.' (AB)

[14] In the clause-incorporating existential, this restriction holds only in the pivot position; definite expressions are allowed in other positions (e.g., as embedded objects), despite also being within the scope of the verb *-qaq* 'have', (i) (see also the definite objects in

the possessee incorporated into *-qaq* 'have' is likewise subject to the Definiteness Restriction.)

(18) **Definiteness Effect on embedded subject**

a. **Taiviti-mik** tikit-tu-qaq-tuq
David-MOD arrive-PART-have-PART.3SG.S
'A person whose name is David arrived.'
#'David arrived.' (AB)

b. ***anaana-nga-nik** tikit-tu-qaq-tuq
mother-POSS.3SG-MOD arrive-PART-have-PART.3SG.S
Intended: 'Her mother arrived.' (AB)

c. ***(pro)*** qai-ju-qa-qqau-nngit-tuq
(3SG.MOD) come-PART-have-REC.PST-NEG-PART.3SG.S
'No one came.'
#'[3SG referent] didn't come.' (IQ)

As seen in (19), the embedded subject of the clause-incorporating existential may also be plural. However, the verb in these examples bears 3SG matrix ϕ-agreement (note that the embedded participial morphology is ϕ-invariant as well). This follows broadly from the generalizations about the case and agreement system sketched in §2.1. For now, we may assume based on the MOD case morphology on the subject that it is inaccessible to the ϕ-probe in the matrix CP-domain.

(19) **No ϕ-agreement with plural existential pivots**

a. arnar-**nik** marruu-**nik** ani-ju-qaq-$\boxed{\text{tuq}}$
woman-PL.MOD two-PL.MOD leave-PART-have-PART.3SG.S
'Two women left.' (AB)

b. Mittimataling-miu-**nik** tikit-tu-qaq-$\boxed{\text{tuq}}$
Mittimatalik-person.from-MOD arrive-PART-have-PART.3SG.S
'People from Mittimatalik (Pond Inlet) arrived.' (AB)

Matrix ϕ-agreement instead indexes what I propose is a null expletive subject in matrix Spec-TP; recall that Inuktitut T^0 has a subject requirement, as alluded to in §2.1. As discussed by Yuan (2018), expletive subjects are canonically indexed by subject ϕ-morphology, but they may also be indexed by object ϕ-morphology under ECM. This is the case in (20a-b), which depict the two existential constructions from (17) embedded under the affixal ECM verb *-qu* 'want'. A partial structure for both examples is provided in (20c).[15]

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examples like (21)). There is also variation within Inuktitut as to whether the affixal verb *-siuq* 'look for' also imposes a Definiteness Restriction on its nominal complement, so for some speakers (ia) cannot contain a referential object. However, for those speakers, this would hold regardless of whether the higher verb *-qaq* 'have' is present.

(i) **No Definiteness Effect on other embedded arguments**

a. (*pro*) **Jaani**-siuq-tu-qa-qqau-juq
(3SG.MOD) John-look.for-PART-have-REC.PST-PART.3SG.S
'Someone was looking for Jaani.' (PI)

b. pani-mik **anaana-mi-nik** iqi-ti-ju-qa-lauq-tuq
daughter-MOD mother-POSS.REFL-MOD hug-AP-PART-have-PART.3SG.S
'A daughter$_i$ hugged her$_i$ mother.' (AB)

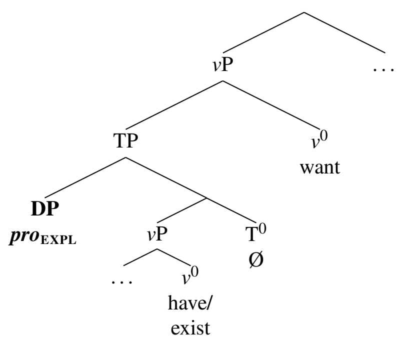
[15] Yuan 2018 also demonstrates that expletive subjects may be antipassivized (i.e., co-occurring with verbal antipassive morphology) when embedded under ECM. This is predicted to be possible with embedded expletive subjects of clause-incorporating existentials as well.

(20) **Expletive subjects of existentials under ECM**

a. [$_{TP}$ (***pro*****EXPL**) inung-mi=luunniit tiki-saali-ju-qa]-qu-nngit-**tara**
(3SG.ABS) person-MOD=NPI arrive-early-PART-have -want-NEG-PART.1SG.S/3SG.O
'I don't want anyone to arrive early.' (AB)

b. [$_{TP}$ (***pro*****EXPL**) Nakusu-up Ilinniaving-mi nutaar-mik ilisaiji-taqa]-qu-**jara**
(3SG.ABS) Nakasuk-GEN School-LOC new-MOD teacher-exist -want-PART.1SG.S/3SG.O
'I want there to be a new teacher at Nakasuk School.' (IG; Yuan 2018, p. 214)

c.



Importantly, these constructions cannot be analyzed in terms of default agreement appearing in the absence of a goal altogether (cf. Preminger, 2011, 2014), since in Inuktitut such configurations result in the loss of agreement morphology altogether, a fact independently shown in Yuan 2023. Rather, the presence of 3SG ϕ-morphology must genuinely reflect a successful Agree dependency established with a 3SG goal. That non-thematic expletives exist at all is, in turn, evidence for a dedicated subject position in Spec-TP. We will revisit this finding in §3.3 below, as this position plays an important role in the embedded clause of the clause-incorporating existential.

3.2 Internal structure of the embedded clause

The examples provided so far have involved simple intransitive verbs and no additional verbal morphology besides participial morphology. However, it can be readily shown that these embedded constituents are syntactically complex, projecting argument structure. The data below demonstrate that the embedded verbs may be logically transitive. Interestingly, however, the resulting constructions are obligatorily antipassive, yielding a MOD-MOD case frame, (21). Put differently, although we have already seen that embedded subjects are necessarily MOD, embedded objects cannot be ABS either. An analysis of these facts will be presented in §5. Note moreover that this generalization extends beyond simple monotransitives, as in (21a-b). In (21c), a causativized verb is antipassivized, resulting in the causee being marked MOD; in (21d), an affixal ECM is antipassivized, so the embedded subject is MOD.[16]

(21) **Incorporated transitive participial clauses are antipassivized**

a. nutaraar-nik **uqumiagar-mik** niri-ju-qa-lau-nngit-tuq
child-PL.MOD candy-MOD buy-PART-have-PST-NEG-PART.3SG.S
'No children bought candy.' (AB)

b. anguti-mik **igalaar-mik** surak-si-ju-qaq-tuq
man-MOD window-MOD break-AP-PART-have-PART.3SG.S
'A man broke the window.' (AB)

[16]See Yuan 2018, ch. 6 for further details on the antipassivization of causatives and affixal ECM verbs. Also, note that the antipassive has several allomorphs, including a null variant (as in (21a)), conditioned by the choice of verb stem, possibly also Aktionsart (Spreng, 2006, 2012).

c. arnar-mik **pani-mi-nik** mumi-tit-si-ju-qaq-tuq
woman-MOD daughter-POSS.REFL-MOD dance-CAUS-AP-PART-have-PART.3SG.S
'A woman$_i$ made her$_i$ daughter dance.' (SN)

d. (*pro*) **uvan-nik** kinguvar-nira-i-ju-qaq-tuq
(3SG.MOD) 1SG-MOD be.late-say-AP-PART-have-PART.3SG.S
'Someone said that I was late.' (AB)

It is also possible for the object to be incorporated into the embedded verb, (22). Although the object is not overtly marked MOD, incorporated nouns in Inuktitut are syntactically akin to antipassive objects, as briefly shown in §2.2 above (e.g., Sadock, 1980; Baker, 1988; van Geenhoven, 1998).[17] This data point thus still adheres to the above generalization.

(22) **Embedded affixal verb with nominal complement**
anguti-mit **nunasiuti**-taaq-tu-qa-lauq-tuq
man-MOD car-get-PART-have-PST-PART.3SG.S
'A man bought a car.' (PG)

The data shown so far have confirmed that incorporated participial constituents contain at least *v*P-level structure. In fact, these elements also demonstrably contain TP. Though T^0 is typically null in these constructions (for reasons to be further discussed in §4.1), the embedded element may also bear overt tense morphology, (23a).[18] Passivization, or A-movement to a derived subject position in Spec-TP, is also permitted inside the incorporated constituent, (23b). Finally, the embedded constituent may also contain sentential negation, (23c), which in Inuktitut heads its own projection above tense (given T < Neg suffix order).

(23) **Incorporated participial clauses may contain TP and NegP**

a. arnar-mik tiki-**qqau**-ju-qa-qqau-juq
woman-MOD arrive-REC.PST-PART-have-REC.PST-PART.3SG.S
'A woman arrived (earlier today).' (AB)

b. igalaar-mik surak-**ta**-**u**-ju-qaq-tuq
window-MOD break-PASS.PART-be-PART-have-PART.3SG.S
'A window was broken.' (AB)

c. inung-mit qai-**nngit**-tu-qaq-tuq
person-MOD come-NEG-PART-have-PART.3SG.S
'Someone didn't come.' (PG)

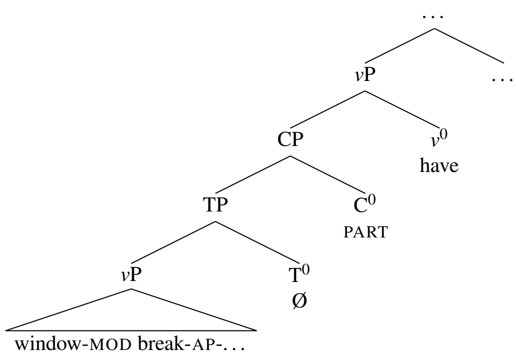
Recall from examples like (14) in §2.2 that Inuktitut has other constructions involving the incorporation of TP clauses. The clauses currently under discussion are larger, given that they also contain participial morphology, realizing C^0. However, notice that there is no ϕ-agreement morphology in the embedded clausal constituent, though we previously concluded that ϕ-agreement is part of the extended CP domain. Although incorporated constituents usually surface without ϕ-morphology, I propose that its absence here is structural: the CP is missing its outermost AgrSP and AgrOP sublayers (further argumentation will be provided in §5).

The tree in (24) illustrates the picture so far, though it omits the embedded subject, the existential pivot. I turn to the exact position of the pivot next.

[17] That being said, Yuan (2025), building on data introduced in Johns 2009 and Beach 2011, demonstrates that incorporated objects in Inuktitut may also behave like ABS objects of ergative constructions (co-occurring with object ϕ-morphology and ERG subjects). This is discussed in §6.1. Thus, the generalization discussed here pertains only to the antipassive incorporation construction.

[18] The double appearance of the REC.PST marker -*qqau* will be explained in §4.1.

(24) **Simplified structure of embedded clause in clause-incorporating existential (based on (21b))**



3.3 The existential pivot controls a clause-internal subject

The existential pivot's syntactic position relative to the embedded participial clause is not immediately deducible based on word order, given that the clause is incorporated. The fact that the nominal exhibits the Definiteness Restriction already shows that it is within the scope of the matrix verb *-qaq* 'have'. But where is it relative to the participial clause, also embedded under *-qaq* 'have'? Based on semantic and morphological diagnostics, I demonstrate that the nominal behaves as though it is simultaneously outside *and* inside of the embedded clause—which I reconcile in terms of *control*. This is an important conclusion, as it informs the structure of the complex constituent under *-qaq* 'have' (see below), the nominal's obligatorily MOD appearance (§5), and the morphosyntactic underpinnings of incorporation (§6).

First, the *clause-external* position of the nominal is evidenced by NPI-licensing. As demonstrated by Bittner (1994) for Kalaallisut, and replicated for Inuktitut by Yuan (2018, 2021, 2022), the disjunctive clitic *=luunniit* 'or' is interpreted as a minimizer under sentential negation (boxed in the examples below). As a baseline, the data in (25) show that NPI licensing is possible in ABS subject position, expected if subjects occupy Spec-TP and negation heads a dedicated projection above TP. Crucially, however, the NPI subject of an incorporated participial clause *cannot* be licensed by embedded negation, (26a), instead requiring matrix negation, (26b).

(25) **Licensing of subject NPI under negation**

a. kina=**luunniit** saqi-lau-nngit-tuq
who.ABS=or show.up-PST-NEG-PART.3SG.S
'Not a single person showed up.' (AB; Yuan 2021, p. 163)

b. atausi=**luunniit** inuk tavvunga-qu-ji-nngit-tuq Jaani-mik
one.ABS=or person.ABS be.here-want-AP-NEG-PART.3SG.S Jaani-MOD
'Not a single person wants Jaani to be here.' (PG)

(26) **Pivot NPI only licensed by matrix negation**

a. *ilinnarti-mik atausir-mig=**luunniit** saqi-nngit-tu-qa-lauq-tuq
student-MOD one-MOD=or show.up-NEG-PART-have-PST-PART.3SG.S
Intended: 'Not a single student showed up.' (AB)

b. ilinnarti-mik atausir-mig=**luunniit** saqi-tu-qa-lau-nngit-tuq
student-MOD one-MOD=or show.up-PART-have-PST-NEG-PART.3SG.S
'Not a single student showed up.' (AB)

In the same vein, embedded indefinite subjects are naturally interpreted by speakers as taking narrow scope under matrix negation but wide scope over embedded negation, as in (27) (see also (30) below). Together, (26)-(27) might indicate that the existential pivot is not located in the canonical subject position (Spec-TP) inside the embedded incorporated clause, since it appears to not be c-commanded by embedded negation.

(27) **Scope of pivot relative to matrix vs. embedded negation**

a. **pani-mik** anaana-mi-nik iqi-ti-ju-qa-lau-nngit-tuq
daughter-MOD mother-POSS.REFL-MOD hug-AP-PART-have-PST-NEG-PART.3SG.S
'No daughter$_i$ hugged her$_i$ mother ($\neg > \exists$).' (AB)

b. **pani-mik** anaana-mi-nik iqi-ti-nngit-tu-qa-lauq-tuq
daughter-MOD mother-POSS.REFL-MOD hug-AP-NEG-PART-have-PST-PART.3SG.S
'One daughter$_i$ didn't hug her$_i$ mother ($\exists > \neg$).' (AB)

On the other hand, there are reasons to think that the existential pivot *is* inside of the embedded clause as well. First, UTAH is an immediate consideration (Chomsky 1981; Baker 1988): as the subject is interpreted as a thematic argument of the embedded verb, it must be first Merged within the extended verb phrase. Second, (27) shows that the embedded subject may bind a reflexive possessor within a lower embedded argument (e.g., the object), as indicated by reflexive agreement on the latter. As further illustrated in (28) below, this reflexive is subject-oriented, in that only external arguments may serve as antecedents (e.g. Bittner, 1994; Manning, 1996). If subject-oriented binding is mediated through a reflexive variety of v^0 (or Voice^0) that Agrees with both the antecedent and the reflexive (e.g. Labelle, 2008; Ahn, 2015; Ershova, 2023), this is more evidence that the antecedent is base-generated in its θ-position inside the embedded clause.

(28) **Binding of reflexive *-mi* by external argument in Spec-*v*P**

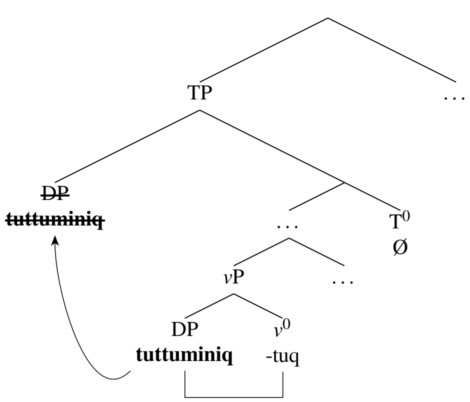
Kaali-p Juuna miiqqa-**min**-nik uqaluttuup-paa
Kaali-ERG Juuna.ABS child-POSS.REFL-MOD tell-IND.3SG.S/3SG.O
'Kaali$_i$ told Juuna$_k$ about his$_i$ children.' (KL; Bittner 1994, p. 173)

Finally, and most strikingly, the subject may in fact appear *overtly* within the embedded clause. To appreciate this point, consider first (29a), which illustrates that affixal verbs may be passivized, an observation first made by Johns (2009). Yuan (2025) demonstrates that incorporated complements of such passivized affixal verbs truly undergo A-movement to Spec-TP. Further details about this construction will be provided in §6.1. Yuan attributes this to an interaction between movement and the Stray Affix Filter (Lasnik, 1981, 1995, et seq.), which prevents morphologically bound elements from being realized without an overt host. As schematized in (29b), the affixal nature of the verb forces the lower copy of an incorporated passivized nominal to be spelled out. General considerations of economy of pronunciation then triggers deletion of the higher copy of movement in Spec-TP (e.g. Landau, 2006).

(29) **Lower copy spell out of passivized subject**

a. **tuttu-miniq**-tuq-ta-u-juq
caribou-former-consume-PASS.PART-be-PART.3SG.S
'The caribou meat is being eaten.' (NB; Johns 2009, p. 195)

b.



With this idea in mind, consider (30). Here, a passivized internal argument is incorporated into an affixal verb, which is, in turn, contained within an incorporated participial clause. Crucially, the passivized nominal is the pivot of the resulting clause-incorporating existential construction. Notice, also, that (30a-b) differ in the locus of negation (matrix vs. embedded), which affects the interpretation of the embedded passivized subject. In line with the generalizations formed with (26) and (27), the nominal still outscopes embedded negation (though not matrix negation), even though it is realized as incorporated into the embedded affixal verb. These constructions, I argue, follow the same derivation as in (29b): morphological well-formedness conditions force the spell-out of the existential pivot as the object of the affixal verb, even though it has been passivized to subject position.

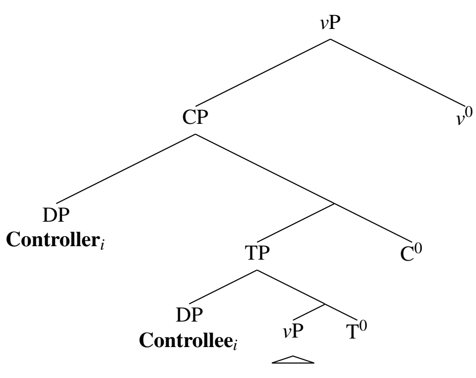
(30) **Clause-internal spell out of subject of clause-incorporating existential**

a. [**sivalaar**-tuq-ta-u-ju]-qa-qqau-nngit-tuq
   cookie-consume-PASS.PART-be-PART -have-REC.PST-NEG-PART.3SG.S
   'No cookies were eaten.' (AB)
   Speaker's comment: "This means all the cookies were not eaten." (¬ > ∃)

b. [**sivalaar**-tuq-ta-u-nngit-tu]-qa-qqau-juq
   cookie-consume-PASS.PART-be-NEG-PART -have-REC.PST-PART.3SG.S
   'A cookie wasn't eaten.' (AB)
   Speaker's comment: "One was left over." (∃ > ¬)

This, again, reifies the impression that the subject is both external and internal to the embedded clause. I propose that this is most straightforwardly captured by a *control structure*, in that the existential pivot is co-indexed with a co-referential element in the embedded subject position, though the pivot itself is located in a higher position. This is schematized in (31), which (foreshadowing §4) depicts the pivot as the specifier of C^0. For convenience, I uniformly depict all nominals as DPs, even though indefinite arguments (like the controller, the existential pivot) are sometimes treated as reduced NPs in the literature.

(31) **Control structure in clause-incorporating existential**



Importantly, while analyses of control classically invoke a null PRO, Inuktitut requires a different representation, since the embedded controlled subject may be forced to surface overtly. This is a *backward control* pattern, in that the controllee rather than the controller is overt (e.g. Polinsky and Potsdam, 2002; Potsdam, 2009; Haddad, 2009; Pietraszko, 2021). In certain languages, such as Telugu, backward control may co-exist with the more familiar forward control pattern, (32). This seems to be the case in Inuktitut, as well, though the two patterns are not optional as they seem to be in Telugu: Inuktitut mostly displays forward control, with backward control emerging only to meet the morphological requirements of affixal verbs.

(32) **Forward and backward control in Telugu**

a. **Kumaar**$_i$ [Δ_i aakali wees-i] saandwic tinnaa-Du
Kumar.NOM hunger.NOM fall-CNP sandwich ate-3M.S
'Having got hungry, Kumar ate a sandwich.'

b. Δ_i [**Kumaar-ki**$_i$ aakali wees-i] saandwic tinnaa-Du
Kumar-DAT hunger.NOM fall-CNP sandwich ate-3M.S
'Having got hungry, Kumar ate a sandwich.' (Haddad, 2009, p. 70)

There are multiple different analyses of backward control. The most prominent analysis invokes A-movement (Hornstein, 1999), meaning that the controllee raises from inside the embedded clause into the higher clause, followed by lower copy spell-out (and concomitant higher copy deletion) (e.g. Polinsky and Potsdam, 2002; Boeckx et al., 2008; Fukuda, 2008; Potsdam, 2009; Haddad, 2009). This seems largely compatible with the Inuktitut facts shown here and would be a simple extension of Yuan's (2025) analysis of the Stray Affix Filter regulating movement chain resolution in incorporation contexts.[19]

* * * *

To summarize this section, we have seen that the clause-incorporating existential consists of an embedded participial clause and a clause-external nominal that serves as the existential pivot. The embedded clause is built up to the (minimal) CP layer but no further. The existential pivot is related to the embedded subject position via control, explaining why it behaves as if it is both external and internal to the clause. The next section develops these conclusions further.

4 A pseudorelative analysis of incorporated participial clauses

Building on the above findings, I now demonstrate that the existential pivot and the embedded clause form a *pseudorelative* construction (Kayne, 1975; Radford, 1977; Cinque, 1992, 1995, a.o.), meaning that they comprise the subject and (clause-sized) predicate of a small clause. Like other pseudorelatives cross-linguistically, the subject is located at the edge of the clause and is tied to the embedded subject position via control. I also contrast these pseudorelative clauses with genuine relative clauses, as this will inform our analysis of complex word-formation in §6.

4.1 Properties of pseudorelatives

The vast majority of the existing research on pseudorelatives has focused on complements of perception verbs in mostly European languages,[20] such as Italian (Cinque, 1992, 1995; Moulton and Grillo, 2016), French (Kayne, 1975; Koopman and Sportiche, 2014), Portuguese (Brito, 1995; Costa et al., 2016), Greek (Angelopoulos, 2015), Spanish (Herbeck, 2020; Aguilar and Grillo, 2021), and more. As shown in (33), both pseudorelatives and subject relative clauses descriptively consist of a subject preceding a clause containing a subject gap. Moreover, they bear surface-identical clausal morphology, such as the complementizer *che*.

[19] That being said, it may also be possible to derive these facts without movement, if we assume that controlled PRO may be overt (e.g. Lee, 2003; Ostrove, to appear) and that the above morphological conditions on chain resolution also determine when deletion under identity is possible (e.g. Saab, 2022). What is not compatible with the Inuktitut facts, however, is a non-movement account of backward control that proposes a single nominal in the embedded clause related to the matrix clause by agreement (e.g. Alexiadou and Anagnostopoulou, 2019; Pietraszko, 2021). Such an analysis cannot account for the morphological underpinnings of nominal spell-out vs. deletion in Inuktitut.

[20] That being said, Cinque 1992 also observes that pseudorelatives are found in existential contexts in Italian, though such constructions are understudied compared to their perception verb complement counterparts.

(33) **Pseudorelative vs. subject relative clause in Italian**

a. Ho visto **Gianni**$_i$ che __$_i$ correva
I.have seen Gianni that run.IMPF
'I saw John run/running.'

b. Vedo **la ragazza**$_i$ che __$_i$ correva
I.see.PRES the girl that run.IMPF
'I see the girl that was running.' (Moulton and Grillo, 2016, pp. 1-2)

However, pseudorelatives and subject relatives are underlyingly distinct constructions, as diagnosable by a constellation of syntactic properties that hold only in the former but not the latter construction. A (non-exhaustive) list of general properties of pseudorelatives is presented below.[21] As we will see, these are also properties of the clause-incorporating existential in Inuktitut.

- The pseudorelative does not denote an individual, but rather a proposition or situation
- The gap in the pseudorelative clause is obligatorily in subject position, reflecting a control structure
- The pseudorelative clause is tense-deficient and must match the tense of the matrix clause
- The case of the subject is determined by the syntactic position of the overall pseudorelative structure

The first two properties listed above have already been established in the paper. The idea that clause-incorporating existentials denote a proposition or situation was already indicated in §3.1; recall that such constructions are translated by speakers as full sentences with indefinite subjects. We also independently arrived at the control analysis in §3.3. In addition to the NPI-licensing and scope facts, recall that the pivot may be construed as the subject of an intransitive, (antipassive) transitive, or passive verb, suggesting a subject-only gap in the embedded clause that is typical of control. The example in (34) below confirms this restriction: the pivot may not be construed as a transitive object.

(34) **No object gap in pseudorelative clause**

*__sivalaar-mik__ niri-ja-<u>qaq</u>-tuq
cookie-MOD eat-PART-have-PART.3SG.S
Intended: 'There is a cookie eaten.' (AB)
(cf. *sivalaaq niri-janga* 'S/he ate the cookie.')

The third property, concerning tense-deficiency, is consistent with the control analysis of pseudorelatives, since controlled elements are restricted to non-finite (i.e., tenseless) clauses cross-linguistically. The incorporated clause in Inuktitut turns out to be tense-deficient as well. Although the example in (35a) was originally introduced in (23a) to illustrate that the embedded constituent contains a TP, I argue that the tense morphology in the incorporated clause is not semantically contentful. In fact, in most examples of the clause-incorporating existential provided in this paper, there is no overt tense morphology in the embedded clause at all; this is further illustrated in (35b), a minimal pair to (35a).

(35) **Embedded tense is semantically vacuous**

a. arnar-mik tiki-**qqau**-ju-<u>qa</u>-**qqau**-juq
woman-MOD arrive-REC.PST-PART-have-REC.PST-PART.3SG.S
'A woman arrived (earlier today).' (AB)

[21] An additional property is that the pseudorelative subject may be a proper name or pronoun, while the head of a (restrictive) subject relative cannot. This comparison cannot be applied to Inuktitut for two reasons. First, recall that the clause incorporation existential construction exhibits the Definiteness Restriction, so such nominals are ruled out to begin with. Second, Compton 2012 argues that all relative clauses in Inuktitut are in fact non-restrictive in nature, meaning that they may be headed by proper names and pronouns.

b. arnar-mik tikit-**Ø**-tu-qa-**qqau**-juq
woman-MOD arrive-PART-have-REC.PST-PART.3SG.S
'A woman arrived (earlier today).' (AB)

Moreover, as shown in (36), genuine tense mismatches are not permitted between the two clauses. Thus, putting (35) and (36) together, I conclude that the embedded clause does not contain independent tense. Rather, embedded T^0 is temporally determined by matrix T^0.

(36) **No independent tense in the embedded clause**

*arnar-mik tiki-**lauq**-tu-qa-**qqau**-juq
woman-MOD arrive-PST-PART-have-REC.PST-PART.3SG.S
Intended possible translations: 'A woman arrived (earlier today)' or 'There was (earlier today) a woman who had arrived in the past.' (AB)

As a possible account of the tense-matching pattern in (35a), I suggest that a matrix and embedded T^0 may enter into a syntactic dependency due to the featural underspecification of embedded T^0, along the lines of existing treatments of sequence of tense (e.g. Zeijlstra, 2012) and voice restructuring (e.g. Wurmbrand and Shimamura, 2017).[22] I also assume that this process is optional, perhaps even dispreferred, given that embedded T^0 is usually null, as in (35b). Abstracting away from further analytical details, the broader point here is that tense-deficiency is another property of pseudorelatives that is borne out in the clause-incorporating existential in Inuktitut.[23]

Finally, the pseudorelative subject receives the case that would be expected based on the syntactic position of the entire pseudorelative constituent. The Italian examples in (37) first provide a point of comparison, demonstrating that whether the subject is NOM or ACC corresponds to whether the pseudorelative is in matrix object or subject position. This also reveals that the subject of the pseudorelative is accessible to morphosyntactic processes such as case assignment arising from the matrix clause.

(37) **Case of subject determined by locus of pseudorelative**

a. Ha visto [**me** che fumavo per strada]
He.has seen me.ACC that smoke.IMPF in street
'He saw me smoking in the street.'

b. [**Io** che fumo per strada] è uno spettacolo che non raccomando
I.NOM that smoke.PRES in street is a sight that not recommend.1SG
'Me smoking in the street is a sight I cannot recommend.' (Moulton and Grillo, 2016, p. 11)

In the clause-incorporating existential in Inuktitut, we have already seen that the pseudorelative subject is necessarily MOD. As alluded to in §2.1 (and further developed in §5 below), this too is due to its syntactic position relative to the matrix clause: its MOD case arises from the pseudorelative constituent's position internal to the matrix *v*P.

To further strengthen the pseudorelative analysis of the incorporated clause, consider now complements of perception verbs in Inuktitut, which may also be plausibly analyzed as pseudorelative CPs. Not only do they too display all of the properties listed above, but they also display the Inuktitut-internal properties already established for incorporated CPs of the clause-incorporating existential. In (38), we see that they too are participial clauses that lack ϕ-agreement with the subject, which is MOD. Therefore, we may conclude that they too are

[22] A more thorough investigation of matrix and embedded tense in Inuktitut is outside of the purview of this paper, but the idea developed here is broadly compatible with Hayashi 2011, the only systematic investigation of Inuktitut tense.

[23] Note that tense-deficiency is not a general property of *all* incorporated clauses in Inuktitut. As illustrated in (14) in §2.2, the TP embedded under the affixal ECM verb *-niraq* 'say' is tensed, as tense mismatches *are* licit in such constructions. This suggests that the tense-deficient nature of Inuktitut pseudorelative CPs is not attributable to independent factors such as being structurally truncated, given that the TP complement of *-niraq* 'say' is smaller, missing the CP layer altogether.

built up to the initial CP layer but no further.[24]

(38) **Perception verb complement clauses are pseudorelative CPs**

a. Kiuru tusaa-juma-juq [uvan-**nik** inngiq-$\boxed{\text{tur}}$-**mik**]
Carol.ABS hear-want-PART.3SG.S 1SG-MOD sing-PART-MOD
'Carol wants to hear me sing.' (AB)
(cf. *uvanga inngiq-tunga* 'I am singing.')

b. Kiuru tusaa-juma-juq [ilitsi-**nik** inngiq-$\boxed{\text{tur}}$-**mik**]
Carol.ABS hear-want-PART.3SG.S 2PL-MOD sing-PART-MOD
'Carol wants to hear you (pl.) sing.' (AB)
(cf. *ilitsi inngiq-tusi* 'You (pl.) are singing.')

Note that, in contrast to incorporated pseudorelative clauses, these standalone clauses bear the same MOD case morphology as their subjects. However, this difference has an independent explanation: recall from §2.2 that incorporated elements in Inuktitut usually do not bear case or other inflectional features. Therefore, since the embedded subject of the clause-incorporating existential is MOD, we may extrapolate based on (38) that the clause itself is underlyingly MOD a s well.

There are additional parallels between these incorporated and standalone clauses. In all elicited examples of the latter, the embedded verb is either intransitive or antipassive; this is consistent with the subject control analysis of pseudorelative clauses. Additionally, the data in (39) demonstrate that the tense matching effects seen above hold for perception verb complements as well, meaning that they are also tense-deficient.

(39) **Perception verb complement clauses are tense deficient**

a. taku-**qqau**-jara [Taami sini-**qqau**-juq]
see-REC.PST-PART.1SG.S/3SG.O Tommy.ABS sleep-REC.PST-PART.ABS
'I saw Tommy sleeping.' (AB)

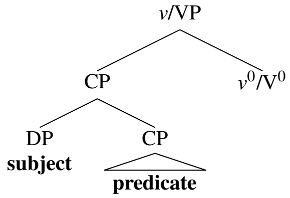
b. *taku-**qqau**-jara [Taami sini-**lauq**-tuq]
see-REC.PST-PART.1SG.S/3SG.O Tommy.ABS sleep-PST-PART.ABS
Intended possible translations: '(Earlier today) I saw Tommy sleeping' or '(Earlier today) I saw Tommy who slept (yesterday).' (AB)

Together, (38) and (39) reify that the case of the pseudorelative subject is indeed determined by the matrix clause. Unlike the incorporated clauses, which must remain *v*P-internal, perception verb complement clauses and their subjects may alternate between MOD and ABS. Again foreshadowing §5, the latter pattern seen in (39) reflects ϕ-Agree with matrix AgrO^0.

Thus, pseudorelative clauses in Inuktitut display several properties that also hold of pseudorelatives in better-studied languages such as Italian. In Inuktitut, these clauses arise as perception verb complements, in line with the cross-linguistic picture, but also in the clause-incorporating existential. As mentioned above, the literature generally agrees that the pseudorelative CP is a predicate of a small clause, whose subject is the nominal related to the CP-internal gap via control. As illustrated in (40), I assume that the small clause subject may Merge directly with the CP predicate as its specifier, following Matushansky 2019, though the data shown here are also compatible with a mediating PredP projection (e.g. Bowers, 1993, 2001; den Dikken, 2006). I additionally suggest that Merging the subject at the edge of the CP is what permits it to be accessible to matrix case and agreement processes in (38)-(39). Finally, this structure may Merge with the affixal verb v^0 *-qaq* 'have' in the clause-incorporating existential or with a perception V^0 (e.g., *tusaa* 'hear' or *taku* 'see').

[24] In addition, pseudorelative complements of perception verbs are canonically taken to encode direct (as opposed to indirect) perception. The idea that these are truncated CPs in Inuktitut dovetails with a similar finding by Grishin (to appear) that direct perception complement clauses are smaller-sized CPs than epistemic complement and root declarative clauses.

(40) **Small clause analysis of pseudorelative**



The small clause analysis advocated for here is similar to other accounts of existentials in which the pivot is the subject of a small clause (e.g. McCloskey, 2014; Irwin, 2018). The relevant point of divergence is that the small clause predicate in Inuktitut is specifically a clause, as opposed to any other non-verbal category.

4.2 Distinguishing pseudorelatives and true relative clauses

Before concluding this section, I briefly demonstrate that the constructions I have analyzed as pseudorelative CPs are syntactically distinct from true relative clauses, despite surface appearances. In Inuktitut (as well as in other Inuit languages), relative clauses also contain participial morphology and bear the same case as the nominals they modify, as shown in (41).[25]

(41) **Relative clauses in Inuktitut**

a. **anguti** [$_{RC}$ nanur-mik kapi-si-juq] ani-juq
man.ABS polar.bear-MOD stab-AP-PART.3SG.S leave-PART.3SG.S
'The man who stabbed the polar bear went out.' (SB; Johns 2007a)

b. Miali kapi-si-juq nanur-**mit** [$_{RC}$ Jaani-up taku-janga]-**nit**
Miali.ABS stab-AP-PART.3SG.S polar.bear-MOD Jaani-ERG see-PART.3SG.S/3SG.O-MOD
'Mary stabs the polar bear that John saw.' (SB; Yuan 2013, p. 63)

However, the two constructions are distinguishable. An initial contrast concerns the observation that relative clauses are not truncated CPs. Whereas we have taken pseudorelative clauses to lack the higher CP-level projections associated with ϕ-agreement, the examples in (41) demonstrate that embedded verbs of relative clauses do bear ϕ-agreement.

These clauses also differ in the locus of the clause-internal gap. In both types of CPs, the gap is necessarily ABS. The Inuit languages are syntactically ergative, so only ABS arguments may be relativized (e.g. Creider, 1978; Johns, 1992; Bittner and Hale, 1996a; Murasugi, 1997). Importantly, what this means is that the gap may be an ABS subject, as in (41a), *or* an ABS object, as in (41b). However, recall from §4.1 above that the incorporated CP of the clause-incorporating existential may only contain a gap in ABS *subject position*. The well-formedness of (41b), in contrast to the ill-formedness of (34), reifies that the latter is not simply a relative clause.

Third, whereas pseudorelative CPs are tense-deficient, genuine relative clauses are not. Relative clauses may bear their own overt tense, distinct from matrix tense, (42):

[25] Here, I abstract away from the possibility, as pursued by Compton 2012, that relative clauses are actually nominal in nature. Specifically, Compton (2012) takes nominals and their relative clause modifiers to involve two DPs in apposition, meaning that one DP is adjoined to the other; this structure will be adopted in §6. If both relative CPs and pseudorelative CPs are subsequently nominalized, this would still be largely compatible with the rest of the paper.

(42) **Relative clauses are not tense-deficient**

a. anguti [$_{RC}$ Jaani taku-**lauq**-tanga sivataaviulauqtu-mik]
man.ABS Jaani.ABS see-PST-PART.3SG.S/3SG.O last.week-MOD
tavva-u-liq-tuq
there-be-PROG-PART.3SG.S
'The man that Jaani saw last week is right there.' (PG)

b. pi-juma-junga [$_{RC}$ niri-**qqau**-jang]-nit ajji-nga-nit
PROFORM-want-PART.1SG.S eat-REC.PST-PART.2SG.S/3SG.O-MOD image-POSS.3SG-MOD
'I want to get the same thing you ate (earlier).' (PI)

A final difference pertains to the incorporation patterns that arise when these constituents are embedded under an affixal verb. The examples first introduced in (17) are repeated as (43) below; we may now analyze the participial clause in (43a) as a pseudorelative clause and its counterpart in (43b) as a relative clause. In (43a), the clause is incorporated and the nominal is stranded; in (43b), the opposite obtains.

(43) **Pseudorelative vs. relative clause in incorporation constructions**

a. **anguti-mik tikit-tu**-<u>qaq</u>-tuq
man-MOD arrive-PART-have-PART.3SG.S
Speaker's translation: 'A man arrived.' (AB)

b. **tikit-tur-mik anguti**-<u>taqaq</u>-tuq
arrive-PART-MOD man-exist-PART.3SG.S
Speaker's translation: 'There is a man who arrived.' (AB)

This asymmetry is shown again in the minimal pair in (44), featuring *-qaq* 'have' in both examples. Incorporating a clause vs. a nominal correlates with an existential vs. possessive reading arising.

(44) **No incorporation of relative clauses**

a. nunasiuti-mik **aupak-tu**-<u>qaq</u>-tuq
car-MOD red-PART-have-PART.3SG.S
'There is a red car.' (AB)
Unavailable: '(S)he has a car that is red.'

b. **aupak-tu**-mik nunasiuti-<u>qaq</u>-tuq
red-PART-MOD car-have-PART.3SG.S
'(S)he has a car that is red.' (AB)
Unavailable: 'There is a red car.'

The contrast in (44), in particular, will be further explored in §6, as it will inform our morphosyntactic analysis of incorporation. For now, it once again confirms that the embedded CP of the clause-incorporating existential is not a relative clause, despite surface appearances.

* * * *

In sum, in this section I have argued that the clause-incorporating existential contains a pseudorelative structure, in that the embedded subject and the incorporated clause are the subject and CP predicate of a small clause. This analysis captures several interlocking properties of these clauses, as they turn out to parallel those of pseudorelative constructions in better-studied languages such as Italian. Additionally, the small clause structure allows us to locate the embedded subject as the specifier of the clausal predicate. In §5 below, we will see how this analysis informs our understanding of the case and agreement system of Inuktitut.

5 Case assignment in incorporated clauses

The generalizations about case and agreement in Inuktitut, first introduced in §2.1 as (10), are repeated below as (45).

(45) **Case and agreement in Inuktitut are tied to the extended CP domain**

a. ERG and ABS cases (indirectly) reflect successful Agree with CP-level ϕ-probes, while MOD case reflects no such ϕ-Agree relations

b. No other functional heads (e.g., located in the TP- and *v*P-domains) are implicated in case assignment and ϕ-agreement

In this section, I demonstrate how these generalizations are evidenced by the case patterns within the clause-incorporating existential, as well as in pseudorelative CPs in Inuktitut more generally. In doing so, I also provide an analysis of these patterns, focusing on the obligatorily intransitive appearance of the embedded clause and the MOD case morphology of the embedded subject.

5.1 Agreement and case at the clausal periphery

The generalizations in (45) are, at least on the surface, compatible with different theories of case assignment. For instance, we could say that ERG and ABS cases are assigned by the same CP-level functional heads also responsible for ϕ-agreement (e.g. Compton, 2017b; Carrier, 2020; Carrier and Compton, to appear). However, in this paper I adopt a slightly more complex analysis, as developed by Yuan (2018, 2022), which takes the connection between case and ϕ-agreement in the clausal periphery to be indirect.

While the former account is simpler and is *prima facie* compatible with the Inuktitut data shown here, a challenge comes from the fact that the subject and object ϕ-probes do not map one-to-one to ERG and ABS case, respectively. This is because ϕ-agreement in Inuktitut generally displays an accusative alignment, rather than an ergative one. As shown in (46), subject ϕ-morphology (in AgrS^0) tracks subject DPs, regardless of whether it is ERG or ABS; also, an ABS DP may be cross-referenced by object ϕ-morphology in AgrO^0 or subject ϕ-agreement in AgrS^0.[26] In other words, ERG and ABS case do not seem to be directly assigned by individual functional heads.

(46) **Ergative vs. accusative case and agreement alignments**

a. **kia** taku-qqau-**va**-$\boxed{\text{uk}}$ $\boxed{\text{Kiuru}}$
who.ERG see-REC.PST-INT.3SG.S-3SG.O Carol.ABS
'Who saw Carol?' (AB)

b. **kina** taku-qqau-**vaa** Kiuru-mik
who.ABS see-REC.PST-INT.3SG.S Carol-MOD
'Who saw Carol?' (AB)

In contrast, an indirect correspondence between agreement and case may accommodate such patterns, while still maintaining that case assignment requires the functional heads responsible for ϕ-agreement. Yuan's account is couched in Dependent Case Theory (Marantz, 1991; Baker, 2015), which canonically takes case to be assigned based on configurational rules, rather than Agreeing heads.[27] For Yuan, the CP-level AgrS^0 and AgrO^0 heads

[26] This holds for most clause type and ϕ-agreement combinations. However, the 'conjunctive' or clause types do display a genuine ergative agreement alignment, in that only ABS subjects and objects may be indexed by ϕ-morphology (ERG subjects are not indexed at all). See Dorais 1988 for the morphological paradigms, and Compton and Yuan to appear for a general overview of agreement alignment in Inuktitut and related languages.

[27] See Poole 2024 and Clem and Deal to appear for Agree-based approaches to dependent case. This could work for Inuktitut as well, so long as the locus of ϕ-Agree remains in the CP-domain. This would be a way of reconciling previous analyses of Inuktitut case as assigned by CP-level heads (Compton, 2017b; Carrier, 2020; Carrier and Compton, to appear) with the evidence for the dependent

are responsible for ϕ-agreement with the subject and object, respectively—but AgrO^0 additionally triggers *movement of the object to Spec-AgrOP* (this account is agnostic about, though is compatible with, AgrS^0 also raising the subject to Spec-AgrSP). As a result, the object c-commands the subject, in accordance with the syntactically ergative nature of the language, as alluded to in §4.2 above (e.g. Creider, 1978; Bittner and Hale, 1996a,b; Murasugi, 1997).[28] Yuan proposes that ERG case is a *downward-oriented dependent case*, assigned to the subject in the presence of a higher c-commanding nominal—the raised object in Spec-AgrOP—and that ABS and MOD are *v*P-external and *v*P-internal 'unmarked' cases, respectively, realized on DPs that do not receive lexical or dependent cases. The latter idea captures the fact that high objects (also indexed by ϕ-agreement) are ABS while low objects are MOD. Presumably, *v*P-internal objects are inaccessible to AgrO^0, resulting in failed Agree in the sense of Preminger 2011, 2014.

Putting everything together, this means that there are two distinct case domains in Inuktitut, *v*P and AgrOP, in which different case assignment rules apply. These rules are summarized in (47).[29]

(47) **Case domains and case rules in Inuktitut**

a. ***v*P domain:** MOD assigned to caseless DPs.

b. **AgrOP domain:** (i) ERG assigned to the lower of two caseless DPs in a c-command relationship; (ii) ABS assigned after (i) to caseless DPs.

As a concrete illustration, consider (48), which depicts a double object construction (repeated from (9a)). First, I assume that the indirect object moves to the edge of the *v*P phase (*pace* Yuan 2022); from there, it is targeted by AgrO^0, which causes it to successive-cyclically move to its final clause-peripheral position. This then feeds dependent ERG case assignment (dashed) to the subject in Spec-TP, which is also indexed by AgrS^0. The indirect object ultimately surfaces as ABS, the *v*P-external unmarked case (boxed), while the direct object receives MOD, the *v*P-internal unmarked case (boxed), since it does not Agree and thus does not move.

(48) **ϕ-Agree and dependent case assignment**

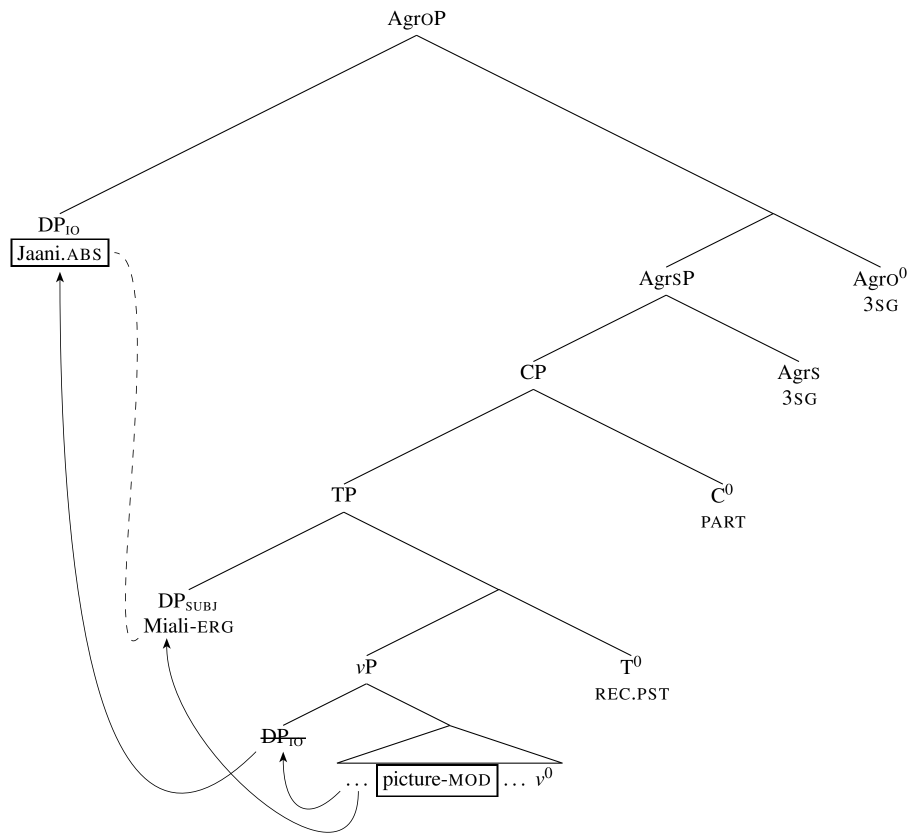
a. Miali-**up** **Jaani** tuni-qqau-janga ajjinnguar-**mik**
Miali-ERG Jaani.ABS give-REC.PST-PART.3SG.S/3SG.O picture-MOD
'Miali gave Jaani a picture.' (IG)

nature of ERG case presented in Yuan 2018, 2022.

[28] In addition to the restriction on relativization discussed in §4.2, there are interpretive asymmetries between ABS (high) and MOD (low) objects, which map roughly to a distinction in specificity or scope (e.g. Bittner, 1994; Beach, 2011; Yuan, 2018, 2022). To account for the surface SOV and SVO word orders often seen in the language, It is plausible that subjects later scramble to an even higher position.

[29] In Yuan 2018, 2021, 2022, the object ϕ-morphology in Inuktitut and other Eastern Canadian varieties is analyzed as the product of *pronominal clitic-doubling*, meaning that this morphology would actually consist of a *raised pronoun* originating from the *in situ* ABS object with which it is co-referential (e.g. Anagnostopoulou, 2006; Baker and Kramer, 2018). It is this raised pronoun that serves as the case competitor for case assignment to the lower subject. In this paper, I abstract away from this particular aspect of Yuan's proposal for simplicity, but it is fully compatible with the facts discussed here.

b.



At this point, the downward-oriented nature of dependent ERG case assignment may seem unorthodox; canonically, dependent ERG is taken to be assigned to the *higher* of two arguments (Marantz, 1991). Moreover, under this more standard approach, object movement to the edge of the *v*P domain is by itself be sufficient to trigger upward dependent ERG assignment to the subject, suggesting that the domain for dependent case assignment is only as large as TP (Baker, 2015). As (48) already depicts this initial movement step, this derivation does not clearly demonstrate why dependent case cannot be assigned until *after* the object raises to Spec-AgrOP. For Yuan 2018, 2022, the evidence for downward ERG case assignment is primarily based on different object movement patterns across Inuit languages. However, I demonstrate below that the clause incorporation existential provides additional evidence for this approach.

5.2 MOD-only nominals in incorporated clauses

Throughout this paper, we have been assuming that the embedded incorporated clause is built up to just the initial CP-layer, so it contains the *v*P structure needed for object shift but not the outermost AgrOP projection proposed here to be responsible for further object movement and *v*P-external dependent ERG case assignment. The initial reasoning for this idea is the observation that there is no ϕ-morphology on the embedded incorporated clause. But, as mentioned in §2.2, incorporated constituents generally surface without inflectional morphology, an observation that previous literature has attributed to the morphology rather than the syntax (Sadock, 1985, 1991; Yuan, 2025). This raises the possibility that the lack of overt ϕ-agreement in embedded incorporated clauses is similarly a purely morphological phenomenon, rather than reflective of missing syntactic structure.

However, the syntactic approach advocated for here makes clear predictions about the case patterns that would arise on the nominals inside the incorporated clause. If the clause is truly structurally truncated, lacking AgrOP in particular, the object would not be able to move past the subject, nor would there be an AgrOP domain for ERG and ABS case assignment to begin with.[30] These predictions, I argue, are borne out by the

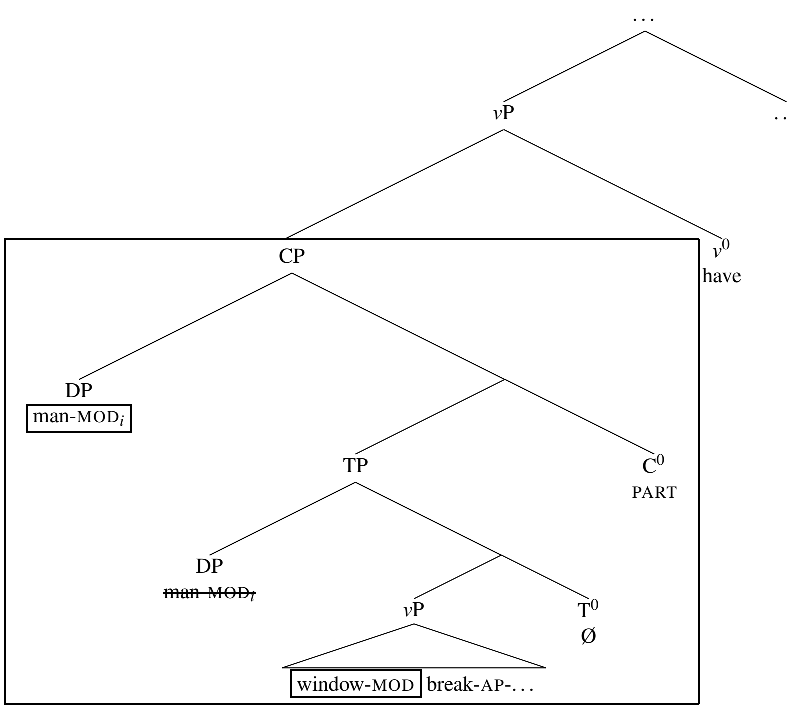
[30] Carrier and Compton to appear independently regard ABS case as a 'root phenomenon', which is broadly compatible with the

fact that embedded arguments in the clause-incorporating existential necessarily bear MOD case, shown again in (49). Following the findings of §§3-4, the embedded subject surfaces in Spec-CP and controls a non-overt embedded subject in Spec-TP. Even though it is located structurally high within the embedded clause, it is not able to surface as ABS due to the lack of AgrOP; rather, it is considered part of the *v*P case domain (headed by *-qaq* 'have') and is accordingly assigned the *v*P-internal unmarked case, MOD. The result is the same for the embedded object, which cannot move to the embedded clausal periphery due to the absence of the AgrOP layer.

(49) **Deriving obligatorily MOD embedded arguments**

a. [$_{CP}$ anguti-**mik** igalaar-**mik** surak-si-ju]-<u>qaq</u>-tuq
man-MOD window-MOD break-AP-PART -have-PART.3SG.S
'A man broke the window.'

b.



On the appearance of verbal antipassive morphology in (49a), recall from §2.1 that the antipassive construction in Inuktitut does not reflect detransitivization. I follow previous literature on Inuktitut that analyzes antipassive morphology as encoding a *v*P-level head that Agrees with a low object (Bok-Bennema 1991; Spreng 2006, 2012; Yuan 2018, cf. Basilico 2019).[31] Put differently, the antipassive arises whenever the object is incapable of escaping the *v*P.

Due to the existential nature of this construction, the incorporated clause and its subject necessarily remain within the scope of v^0 *-qaq* 'have', so the subject is obligatorily MOD. Recall from §4.1, however, that in perception verb constructions both MOD and ABS are available; a clearer minimal pair is provided in (50). This alternation, too, follows from the present analysis: in non-existential contexts, the pseudorelative constituent

analysis pursued here, in that root clauses are not syntactically reduced.

[31] To be precise, these authors suggest that this Agree dependency is results in MOD case assignment. Given that the current analysis takes MOD to be a *v*P-internal unmarked case, I suggest instead that antipassive morphology simply functions as a morphological signal that the *v*P *contains* a low object, whereas its absence indicates failed Agree due to the *v*P being empty of its arguments. In this recharacterization, the antipassive-bearing head is not responsible for MOD case assignment, though its presence is correlated with the nominal being assigned MOD. This is effectively the analysis of Halpert 2012, 2015 for a similar *v*P-level morphological alternation in the Bantu language Zulu. In contrast, Basilico 2019 proposes that the antipassive in Inuit-Yupik languages introduces the MOD-marked argument, but this is not compatible with findings of Yuan 2018 that non-thematic expletive subjects may be antipassivized, briefly noted in fn. 15).

may optionally remain *v*P-internal or move to Spec-AgrOP.

(50) **Perception verb pseudorelatives may be MOD or ABS**

a. tautu-qqau-nngit-tunga [Kiuru-**mik** inngiq-tur-**mik**]
look.at-REC.PST-NEG-PART.1SG.S Carol-MOD sing-PART-MOD
'I didn't look at Carol singing.' (AB)

b. tautu-qqau-nngit-**tara** [**Kiuru** inngiq-**tuq**]
look.at-REC.PST-NEG-PART.1SG.S/3SG.O Carol.ABS sing-PART.ABS
'I didn't look at Carol singing.' (AB)

As yet another piece of evidence for this analysis, we may contrast the clause-incorporating existential with another construction that, strikingly, *does* permit incorporation of a full clause built up to AgrOP. In (51), we see that the affixal verb *-la* 'say' embeds full participial clauses with subject and object ϕ-agreement. The resulting biclausal constructions are indeed single prosodic words, as evidenced by morphophonological changes affecting the final segments of the incorporated verbal stem (Beach 2011, pp. 376-379; see also Bottineau et al. 2010 for related discussion).[32] Crucially, the embedded nominal arguments in these constructions display the same case patterns as they would in root clauses. For instance, the overt object in (51a) and transitive subject in (51b) are ABS and ERG, respectively, rather than MOD; in (51c) the embedded nominals follow an ABS-MOD antipassive case frame.

(51) **Incorporation of extended CP retains embedded agreement and case**

a. (*pro*) [$_{AgrOP}$ **(*pro*)** **Jaani** malit-**tangaa**]-la-laur-tunga
(1SG.ABS) (3SG.ERG) John.ABS follow-PART.3SG.S/3SG.O -say-PST-PART.1SG.S
'I said that s/he was following John.' (cf. *Jaani malit-tanga* 'S/he is following John.')

b. (*pro*) [$_{AgrOP}$ **Jaani-up** **(*pro*)** malit-**tangaa**]-la-laur-tunga
(1SG.ABS) John-ERG (3SG.ABS) follow-PART.3SG.S/3SG.O -say-PST-PART.1SG.S
'I said that John was following him/her.' (cf. *Jaani-up malit-tanga* 'John is following him/her.')

[32] According to Beach 2011, it is not clear whether the embedded element reflects direct or indirect quotation. Beach provides (ia), which could point towards a direct quotation analysis (though it is also compatible with an indexical shift analysis, with the caveat that the latter is not otherwise known to occur in Inuktitut). On the other hand, examples like (51c) and (ib) below demonstrate that the embedded elements may be discontinuous, which is not expected for true quotations.

(i) **Further examples of full clause incorporation into *-la* 'say'**

a. nagli-gusuk-tungaa-la-laur-tuq
love-AP-PART.1SG.S-say-PST-PART.3SG.S
'He/she$_i$ said that he/she$_i$ loves someone.'
Unvailable: 'He/she said that I love someone.' (SB; Beach 2011, p. 380)

b. nagli-gusuk-tungaa-la-laur-tunga Mary-mit
love-AP-PART.1SG.S-say-PST-PART.1SG.S Mary-MOD
'I said, "I love Mary."' or 'I said that I love Mary.' (SB; Beach 2011, p. 377)

Note also that Beach 2011 adopts a different morphological decomposition, *-Vla* 'say', than the one assumed here, in that the affixal verb begins with an underspecified vowel whose identity must match the final vowel of the stem. In this paper, I follow Bottineau et al. 2010 in taking the verb form to be *-la*. Under this approach, the morphophonological changes apply purely to the incorporated stem and not the verb itself. If the stem ends in a consonant, it is deleted, a process that applies regularly with suffixation in the Inuit languages, and the final (remaining) vowel is lengthened. Finally, *la* 'say' is sometimes also represented as non-affixal (e.g. Johnson and Allen, 2022; Lee and Allen, 2023), though this representation does not capture the morphophonological effects seen in (51); moreover, the data given in these sources show the embedded clause immediately linearly preceding the higher verb, so they are still compatible with an affixal analysis.

c. (*pro*) $[_{AgrOP}$ ***(pro)*** **Mary=lu** malit-**tuguu**]-la-laur-tunga $[_{AgrOP}$ **tuktu-mit**]
(1SG.ABS) (1SG.ABS) Mary.ABS=CONJ follow-PART.1PL.S -say-PST-PART.1SG.S caribou-MOD
'I said that Mary and I were following the caribou.' (cf. *Mary=lu malit-tugut tuktu-mit* 'Mary and I are following the caribou.')
(SB; Beach 2011, pp. 378-379)

This further reinforces the analysis of case assignment developed above; the availability of these cases is predicted if AgrOP is projected in the embedded clause. Additionally, (51) teaches us that not all clauses that undergo incorporation are truncated; this is an important point that will inform our analysis of incorporation and complex word-formation in §6 below.

* * * *

This section has offered a detailed analysis of agreement and case in Inuktitut, building on the generalizations first introduced in §2.1. The clause-incorporating existential provides novel evidence that ERG and ABS cases are found only in clauses built up to AgrOP; their absence in the embedded CP of the clause-incorporating existential is due to its truncated status.

More broadly, the discussion above has revealed that incorporated and standalone clauses need not differ structurally. Embedded clauses in clause-incorporating existentials are pseudorelative CPs, as are standalone complement clauses of perception verbs; moreover, full AgrOP clauses that are morphosyntactically identical to root declaratives may also undergo incorporation, as long as a higher affixal verb selects for them.

6 Clause incorporation as a window into complex word-formation

Having established the morphosyntactic structure of the clause-incorporating existential construction, we now turn to the grammatical underpinnings of clause incorporation and its broader implications for complex word-formation in Inuktitut. I posit that we may account for all incorporation phenomena in Inuktitut, nominal and clausal, under a unified account of complex word-formation: incorporation takes place whenever a constituent (of any category) serves as the syntactic complement of a verb morphologically subcategorized as affixal. Building on Yuan 2025 (which, in turn, adopts key insights of Sadock 1985, 1991), this may be modeled using the postsyntactic operation of Merger, which creates complex words from structurally adjacent heads (e.g. Bobaljik, 2002; Harizanov and Gribanova, 2019).

This idea can be contrasted against the theory of polysynthetic word-formation developed by Compton and Pittman (2010), which eschews the notion that individual morphemes may be specified as affixal or not; rather, in polysynthetic languages like Inuktitut, syntactic constituents of certain categories are mapped directly to phonological words. Importantly, the clause-incorporating existential offers novel evidence against this theory, as it requires that morphologically bound (i.e., incorporated) elements always be syntactically smaller than their standalone counterparts. The present Merger-based account makes the correct empirical predictions for Inuktitut, albeit leaving the language's polysynthetic status largely unexplained.

6.1 A postsyntactic analysis of incorporation

Although incorporated nominals are usually analyzed as structurally smaller than their independent counterparts, both in Inuit languages (e.g. Bok-Bennema and Groos, 1988; Bittner, 1994; van Geenhoven, 1998; Johns, 2007b; Compton and Pittman, 2010) and cross-linguistically (e.g. Baker, 1988), Yuan (2025) proposes that, at least in certain Inuktitut varieties, incorporated and non-incorporated nominals are in fact syntactically equivalent, despite surface appearances (see also Sadock 1985, 1991).

The key evidence for this idea comes from a series of observations revealing that incorporated nouns in Inuktitut are syntactically active. As discussed in §2.2, incorporated objects in the Inuit languages are generally considered to be underlying MOD, akin to antipassive objects (see also fn. 17 in §3.2). However, as Yuan shows, incorporated nominals in Inuktitut may also exhibit properties otherwise characteristic of standalone ABS DPs (see also Johns 2009 and Beach 2011). In (52), the incorporated nominal is passivized; additionally, (52a) shows that modifiers of passivized incorporated nominals show ABS case concord, and (52b) reveals that passivization allows the noun to bind into the oblique agent. Therefore, as also briefly alluded to in §3.3, passivized incorporated nominals genuinely undergo A-movement.

(52) **Passivized incorporated nominals in Inuktitut**

a. **pingasut** **ujami**-<u>liuq</u>-ta-u-**jut**
three.ABS necklace-make-PASS.PART-be-PART.3PL.S
'Three necklaces are being made.' (CH; Yuan 2025, p. 526)

b. **aasivar**-<u>tuq</u>-ta-u-juq nuliaqta-**mi**-nut
spider-consume-PASS.PART-be-PART.3SG.S mate-POSS.REFL-ALLAT
'The spider$_i$ is being eaten by its$_i$ mate.' (AB; Yuan 2025, p. 527)

In (53a), the incorporated object is indexed by object ϕ-morphology, co-occurs with an ERG subject, and is interpreted as though it scopes over sentential negation—all of which is expected if it moves to the clausal periphery, per the analysis of ergative constructions from §5.1. Moreover, compare this construction with the more familiar-looking antipassive incorporation construction in (53b). Thus, incorporation constructions exhibit ergative vs. antipassive alternations, just like other transitive constructions do.

(53) **Ergative vs. antipassive alternations in incorporation constructions**

a. *Felicitous context provided by speaker:* "Ulak won't eat salmon if there is Arctic char around."
Ula-**up** **iqalu**-<u>tu</u>-runna-nngit-**tanga**
Ulak-ERG fish-consume-MODAL-NEG-PART.3SG.S/3SG.O
'Ulak won't eat a particular fish.' ($\exists > \diamond/\neg$)

b. *Felicitous context provided by speaker:* "Ulak has a seafood allergy."
Ulak **iqalu**-<u>tu</u>-runna-nngit-**tuq**
Ulak.ABS fish-consume-MODAL-NEG-PART.3SG.S
'Ulak can't/won't eat (any) fish.' ($\diamond/\neg > \exists$) (CH; Yuan 2025, p. 540)

Yuan additionally notes that these facts are not unique to Inuktitut. Rather, syntactically active incorporated nominals may be a feature of languages with obligatory incorporation (e.g. Wojdak, 2008; van Urk, 2020). If incorporation is driven by grammatical properties that have nothing to do with the nominals themselves, then it should not matter if these nominals are structurally reduced or not.

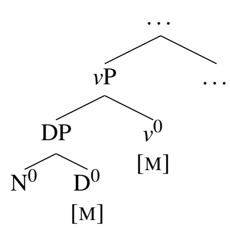
Based on this logic, Yuan proposes that Inuktitut nominal arguments are uniformly DPs, regardless of whether they undergo incorporation, while incorporation is triggered by the morphological subcategorization properties of the verbs selecting for them. Incorporation is a postsyntactic process, meaning it takes place after any syntactic operations that may apply to the nominals.[33] The nominals may therefore participate in ϕ-agreement, case assignment, and movement processes, accounting for the data in (52)-(53). However, if they happen to be embedded under a verb that is specified as affixal, the Stray Affix Filter of Lasnik 1981, 1995 forces the spell-out of the movement copy adjacent to the verb (as already shown in (51c) in §3.1). This results in the appearance of syntactically active incorporated nouns.

[33]Here, Yuan 2025 diverges from the proposal of Sadock (1985, 1991), which likewise takes incorporation to be morphological and thus independent of the syntax. However, Sadock's account is couched in his Autolexical Theory, in which syntax and morphology operate separately but in tandem. In contrast, in the Distributed Morphology framework (Halle and Marantz, 1993) that Yuan assumes, the output of the syntactic module is what the morphology operates on.

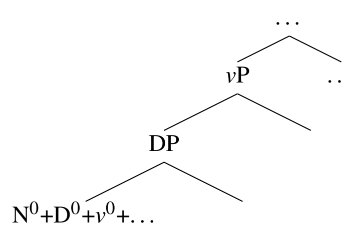
The postsyntactic word-formation operation invoked by Yuan is (Morphological) Merger (e.g. Marantz, 1988; Embick and Noyer, 2001; Harizanov and Gribanova, 2019; Snigaroff, 2024), which creates a complex head out of two adjacent heads and realizes these elements as morphologically bound, as shown in (54). In noun incorporation constructions, v^0 undergoes Merger with the head of its DP complement; D^0 also undergoes Merger with the head of its own complement, etc. Formalizing this idea further, I adopt the notation introduced in Harizanov and Gribanova 2019: heads that are lexically specified as having to undergo Merger with an adjacent element bear a feature [M].[34] Note that (54) omits the functional heads above v^0, which are also affixal and thus should also be specified as bearing [M].

(54) **Accounting for noun incorporation of DPs with Merger**

a. Syntax:



b. PF (after Merger):



An important detail of this analysis is that Merger applies between *structurally adjacent* heads, rather than linearly adjacent ones, accounting for the observation that only *complements* may undergo incorporation. In contrast, adjuncts and specifiers never incorporate, even when they too are within the scope of an affixal verb. Following the distinction from Embick and Noyer 2001, this means that the subtype of Merger invoked here is specifically *Lowering*, which is sensitive to hierarchical order and may thus ignore intervening material such as adjuncts (as opposed to Local Dislocation, which operates on linearly adjacent elements, regardless of their syntactic status). In (55), for instance, the nominal *guulu* 'gold', which may normally incorporate when serving as a direct object, can no longer do so when serving as the modifier of another nominal.[35]

(55) **No incorporation of adjuncts**

a. **guulu**-<u>taa</u>-ruma-junga
gold-get-want-PART.1SG.S
'I want to get some gold.'

b. **guulu-mik** ujami-<u>taa</u>-ruma-junga
gold-MOD necklace-get-want-PART.1SG.S
'I want to get a gold necklace.'

c. *ujami-mik **guulu**-<u>taa</u>-ruma-junga
necklace-MOD gold-get-want-PART.1SG.S
Intended: 'I want to get a gold necklace.' (IQ; Yuan 2025, p. 531)

The idea that Merger requires first establishing head-complement relations in the syntax is crucial to our analysis of clause incorporation, as it will allow us to distinguish between the incorporation patterns of pseudorelative clauses and genuine relative clauses. I turn to this next.

[34] As noted by Yuan (2025, p. 529) it does not matter whether Merger proceeds upward like traditional head movement or downward. In (54b), the illustration of Merger as proceeding downward is arbitrary and does not reflect any deeper analytical decisions. In Harizanov and Gribanova 2019, the directionality of Merger is distinguished by the +/– feature values (e.g., [M:+]) and [M:–]); I omit these values in (54b) for convenience.

[35] Yuan 2025 additionally suggests that there may be variation within Inuit languages as to which subtype of Merger is relevant. For instance, there are other Inuktitut varieties in which adjuncts may undergo incorporation; see also Snigaroff 2024 for a similar conclusion based on Western Aleut (distantly related to the Inuit languages). Under the present analysis, these languages would derive complex word-formation using Local Dislocation.

6.2 Extension to clause incorporation

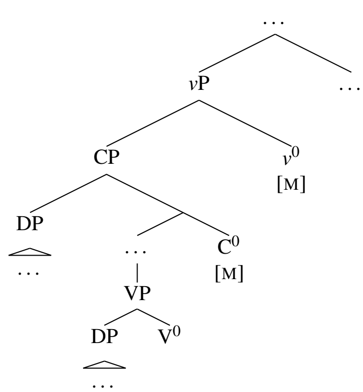
The above analysis of Yuan 2025 focuses only on the Merger between v^0 and the head of its nominal complement (e.g., D^0). Importantly, this idea may be readily extended to account for the incorporation of clausal constituents. I illustrate this below with the clause-incorporating existential, but the same logic holds for incorporated TP clauses such as (14), as well as incorporated AgrOP clauses such as (51).

Consider an example such as (21b), repeated as (56a) below. The (simplified) syntax of this construction is represented as (56b). If v^0 is morphologically subcategorized to undergo Merger (i.e., bears [M]), regardless of the syntactic category of its complement, then in the clause-incorporating existential construction the v^0 *-qaq* 'have' should undergo Merger with embedded C^0; as established in the sections above, the embedded CP is v^0's complement. Since C^0, T^0, and other functional heads along the clausal spine are all affixal as well, they too bear [M], causing Merger to apply iteratively between each pair of structurally adjacent heads. Since the embedded verb does not incorporate its complement, it does not bear [M] so Merger does not proceed further.[36] The result is a complex verb consisting of several functional heads along the matrix and embedded clausal spines, with both nominal arguments surfacing as separate words, (56c).

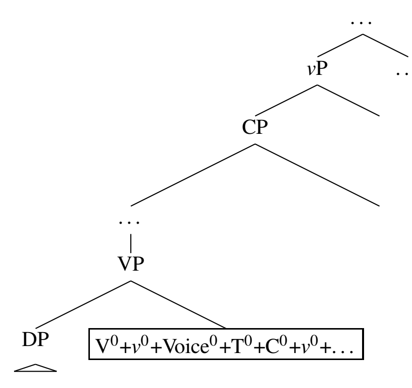
(56) **Deriving clause incorporation**

a. anguti-mik igalaar-mik surak-si-ju-qaq-tuq
man-MOD window-MOD break-AP-PART-have-PART.3SG.S
'A man broke the window.' (AB)

b. Syntax:



c. PF (after Merger):



This account also explains why the nominal pivot is never incorporated in this construction, even though it is within the scope of the matrix verb. Most generative analyses of incorporation, whether syntactic or postsyntactic (e.g., Baker 1988, 2009; Baker et al. 2005; Johns 2007b, 2009; Barrie and Mathieu 2016), adhere to standard locality conditions, meaning that the highest accessible element in v^0's c-command domain should incorporate. According to our pseudorelative analysis from §4, the nominal is located in Spec-CP. As it is sometimes assumed that heads and their specifiers are equidistant (e.g. Pesetsky and Torrego, 2001; Hornstein, 2009; Oxford, 2024), it is not immediately obvious on locality grounds alone why v^0 should not undergo Merger with the D^0 of the pseudorelative subject instead of embedded C^0. However, since the present approach explicitly operates on heads of complements, D^0 is not a viable target by due to the DP's status as a *specifier*.

Finally, along the same lines, this distinction allows us to derive the contrasting incorporation patterns of pseudorelative vs. genuine relative clauses; recall from §4.2 that the latter never incorporate. Relative

[36]Following Johns's (2007b) position that affixal and non-affixal verbs are v^0s and V^0s, respectively, we could tentatively generalize that in Inuktitut functional heads bear [M] while lexical heads do not. Alternatively, we may assume that the morphological subcategorization requirement of any given head is completely lexically determined, that is, not predictable by syntactic category (this is possibly more in line with the discussion in Harizanov and Gribanova 2019, pp. 486-487, but they do not consider polysynthetic languages such as Inukitut). I leave a more detailed consideration of the determinants of affixal vs. standalone elements of Inuktitut for future work.

clause CPs are adjuncts within DPs, so they are excluded from incorporation for the same reason as other non-complements.

Thus, to summarize, I have shown that the postsyntactic treatment of noun incorporation of Yuan 2025 may straightforwardly capture clause incorporation as well. If an affixal verb takes a CP as its complement as opposed to a DP, it would undergo Merger with C^0. As C^0, T^0, and other heads along the clausal spine are also specified as affixal, the end result is a complex verb spanning multiple clauses.

6.3 Comparison with Compton and Pittman 2010

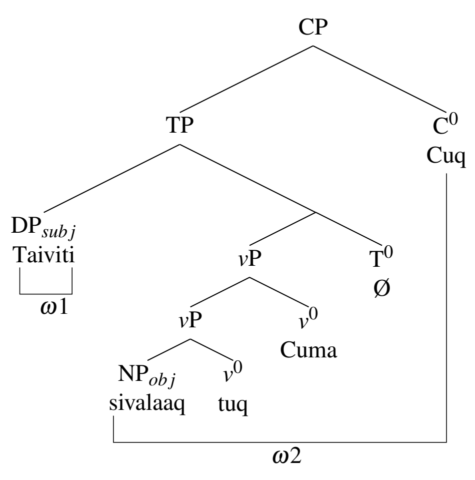
Finally, I contrast our Merger-based approach with the influential theory of complex word-formation developed by Compton and Pittman (2010) for the Inuit languages (and subsequently extended to other polysynthetic languages; see Bliss 2013; Barrie and Mathieu 2016; Ershova 2020). According to this account, these languages map syntactic phases to single phonological words, and, correspondingly, subphasal elements to subword morphology. Compton and Pittman moreover posit that only CPs and DPs are phases; transitive *v*Ps and other elements commonly identified as phasal (e.g., categorizing heads) are crucially not.[37]

To account for incorporation, Compton and Pittman propose that incorporated nouns are reduced NPs rather than DPs, and additionally observe that incorporated verbal constituents, as in the ECM constructions exemplified in (14), are TPs at most. Consider how this analysis captures a sentence as in (57), which contains a standalone subject and an incorporated object.[38] The subject is a DP, so it is spelled out as an independent word. Then, once the CP phase is built, the remainder of the clause is also spelled out. Because the object is by assumption smaller than DP, it is spelled out with the verbal elements within this phase—resulting in its incorporated appearance.

(57) **Phases as phonological words in Inuktitut**

a. Taiviti **sivalaar**-<u>tu</u>-ruma-juq
David.ABS cookie-consume-want-PART.3SG.S
'David wants to eat the cookie/cookies.' (AB)

b.



Put differently, under this account, there need not be any syntactic or morphological differences between what we have been referring to as affixal vs. non-affixal verbs, since the (non-)occurrence of incorporation is fully predicted by the size of the incorporated element. This treatment of word-formation thus makes a strong

[37]This view is therefore incompatible with proposals for typologically similar languages as containing word-internal phases (e.g. Newell and Piggott, 2014).

[38]Recall that modifiers of incorporated nominals are stranded. This follows from Compton and Pittman's (2010) analysis if they are CPs (adjectival modifiers bear participial morphology, like relative clauses). It is also possible that these modifiers are nominalized, i.e., DPs, as proposed by Compton (2012).

prediction: all incorporated (non-phasal) elements are structurally smaller than their standalone (phasal) counterparts. In other words, there cannot be some XP that is both able to undergo incorporation and surface as an independent word, depending on the context. However, I argue below that this prediction is too strong and undergenerates.

For noun incorporation, Compton and Pittman's treatment of such nouns as NPs rather than DPs is motivated by their morphologically uninflected appearance, as well as by the assumption that they are syntactically inert, unable to be targeted for ϕ-agreement and movement operations. But, as discussed above, while the absence of inflectional morphology on incorporated nominals could be attributed to missing syntactic structure, it could also be a purely morphological matter (as proposed by Sadock 1985, 1991 and Yuan 2025). Furthermore, under certain affixal verbs, nominals may in fact appear with inflectional morphology indicating the presence of a DP layer. This is the case with nominals incorporated into the copula *-u* 'be', as shown in (58) with an incorporated object bearing possessive agreement morphology.[39]

(58) **Inflected incorporated object with the affixal verb *-u* 'be'**

Kiuru **angaju-ngi**-<u>u</u>-quuji-juq
Carol.ABS elder-POSS.3PL-be-seem-PART.3SG.S
'Carol resembles her elder relatives.' (AB)

Finally, as discussed in §6.1 above, incorporated nominals in Inuktitut behave syntactically like (standalone) DPs even when they are morphologically uninflected, suggesting that they are in fact underlyingly DPs. If so, then we do not expect them to ever undergo incorporation under Compton and Pittman's analysis.

Importantly, the same counterargument may be made about incorporated clauses. Compton and Pittman do not consider full vs. truncated CP-sized clauses (they compare only CPs and TPs). However, given our proposed distinction between full AgrOP clauses and reduced CP clauses, we may ask how these would fit under a phase-based analysis. If the highest extended projection of the CP domain corresponds to the CP phase, this would correspond to AgrOP; truncated CPs would, in turn, be considered subphasal. This means that full AgrOPs should always be mapped to standalone words, while clauses missing these outermost layers should never be.

However, the data shown in this paper demonstrate that this prediction is not borne out. Reduced CPs are indeed incorporated in the clause-incorporating existential construction, but the same pseudorelative clauses form independent words when they serve as complements of perception verbs, shown again in (59a). In a similar vein, full AgrOPs are typically realized independent words, but, as seen in §5.2, under the right conditions (e.g., embedded under an affixal verb selecting for AgrOP complements) they may be realized as incorporated as well, repeated in (59b).

[39] Compton and Pittman (2010) do not discuss the pattern in (58), but do acknowledge the availability of nominals bearing possessive and case morphology under affixal verbs encoding location, as in (i) (fn. 20, p. 2174). They follow Sadock 2002 in treating such verbs as enclitic rather truly affixal (given Sadock's observation that they may optionally surface as discontinuous from their objects), and suggest accordingly that these data are outside the scope of their analysis. However, this cannot be extended to *-u* 'be', which does not display any clitic-like properties.

(i) **Incorporation of inflected objects under verbs of location**

a. **illu-ga**-<u>no</u>-vunga
house-POSS.1SG-go.to-IND.1SG.S
'I'm going to my house.' (Compton and Pittman 2010, p. 2174, citing Johns 2007b)

b. **Nuum-mi**-<u>ip</u>-punga
Nuuk-LOC-be.in-IND.1SG.S
'I am in Nuuk.' (Compton and Pittman 2010, p. 2174, citing Sadock 2002)

(59) **Standalone truncated clauses and incorporated full clauses**

a. taku-qqau-jara Taami **sini-qqau-juq**
see-REC.PST-PART.1SG.S/3SG.O Tommy.ABS sleep-REC.PST-PART.ABS
'I saw Tommy sleeping.' (AB)

b. (*pro*) Jaani-up (*pro*) **malit-tangaa**-la-laur-tunga
(1SG.ABS) John-ERG (3SG.ABS) follow-PART.3SG.S/3SG.O-say-PST-PART.1SG.S
'I said that John was following him/her.' (SB; Beach 2011, pp. 378)

In contrast, the Merger-based analysis adopted in this paper does correctly predict that given constituent may in principle be incorporated or standalone, depending on morphosyntactic context. In (59b), for instance, the full AgrOP clause happens to be the complement of an affixal verb, whereas most other verbs selecting for an AgrOP clause are not affixal.

Zooming out, a broader difference between these accounts pertains to their treatment of polysynthetic languages as a defineable class. Compton and Pittman (2010) do not explicitly cast their analysis as a general framework for polysynthetic word-formation. However, in suggesting that "these languages apply a different level of the prosodic hierarchy to syntactic structure (i.e., phases are mapped to phonological words, as opposed to phonological phrases, as may be the case in other languages)" (p. 2189), it is implied that polysynthetic vs. non-polysynthetic languages may arise from parametric variation in syntax-prosody mappings. In contrast, the Merger operation invoked above is not specific to polysynthetic languages: Inuktitut appears polysynthetic due to *most* heads in the language being morphologically specified as affixal (i.e., bearing [M]), but otherwise there is nothing that sets a language like Inuktitut apart from a non-agglutinating language like English. The Merger-based approach thus does not make any strong predictions about morphological typology.

Although the lack of predictive power could be viewed as a downside, it is also advantageous for other reasons. First, the idea that a single mechanism may capture such a diverse range of morphological patterns results in a more parsimonious theory of complex word-formation overall. Second, although prior work has sought to characterize polysynthetic languages as a unified class (e.g. Baker, 1996), there is no consensus around this idea, as other research has argued that polysynthesis should simply be viewed as a typological label without deeper theoretical status (e.g., Evans and Sasse 2004; see also niga 2019 and Compton and Bliss 2023 for recent overviews of different views). Whether this is the right direction overall requires a deeper evaluation of both empirical coverage and broader theoretical considerations, to be left for future work on this topic.

7 Conclusion

In this paper, I have investigated the (otherwise underexplored) morphosyntax of the clause-incorporating existential in Inuktitut. I have argued that the incorporated clause in this construction is a pseudorelative small clause predicate, whose nominal subject is the pivot of the existential. I have also demonstrated that the incorporated clause is a reduced CP and have outlined the implications of this structure for ERG and ABS case assignment. Finally, the fact that the embedded clause, rather than the subject, undergoes incorporation is derived from a postsyntactic theory of word-formation, whereby complex word formation is determined by the morphological subcategorization properties of individual heads, rather than syntactic size alone.

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